

Lecture 6: Unsupervised Learning II
Probability and Variational Autoencoders
Conditional and Disentangled Generation
Digits, Blood Cells, Ising Configurations, Galaxies

Deep Learning for Scientific Discovery



Marcin Płodzień

Atomic Optics Department
Institute of Theoretical Physics
Jagiellonian University in Kraków

Recap: Understanding Data without Labels

- PCA and t-SNE: a few coordinates that describe each sample
- k-means and DBSCAN: groups of similar samples
- autoencoders: a learned compression into a latent space; the reconstruction error flags unusual samples
- anomaly detection: unusual blood cells and credit-card fraud flagged by a large reconstruction error

Today: learn the distribution of the data itself, and draw new samples from it.

1

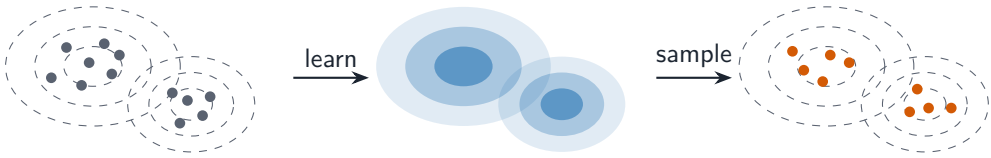
Generative Models

JAGIELLONIAN UNIVERSITY
IN KRAKOW

Learning a Probability Distribution

We have only data $\mathbf{x}_1, \ldots, \mathbf{x}_n$: images, spectra, spin configurations. No labels, nothing else. We assume they are independent samples of an unknown probability distribution $p(\mathbf{x})$.

Goal: a model $p_\theta(\mathbf{x})$ that we can *sample*: new $\mathbf{x} \sim p_\theta$ that look as if they came from $p(\mathbf{x})$ itself, with the same statistics (shapes, frequencies, correlations, energies), and that are not copies of the data.



data: samples of an unknown $p(\mathbf{x})$ (dashed)

model $p_\theta(\mathbf{x})$

new samples: where $p(\mathbf{x})$ puts its data

Supervised learning models $p(y \mid \mathbf{x})$; Lecture 5 found structure in the data; today we learn $p(\mathbf{x})$ itself.

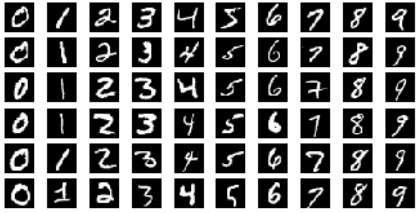
Generating New Data

A **generative model**

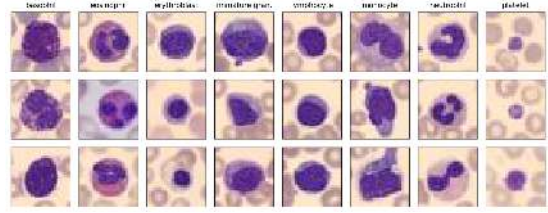
- learns an approximation $p_\theta(\mathbf{x})$ of $p(\mathbf{x})$ from the samples (θ: its parameters), by making the observed data probable under the model
- draws **new samples**: instances not in the data set, but that could have been
- can be checked: the generated samples must reproduce the statistics of the data (for spins, the physics: magnetisation, energy, correlations)

In science: new molecules and materials, protein structures, weather ensembles, fast surrogates of expensive simulations; and the learned p_θ itself as a description of the data.

The Data of This Lecture



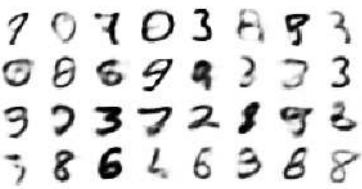
MNIST (Lecture 3): 70 000 handwritten digits 0–9, 60 000 for training and 10 000 for testing; $\mathbf{x}_i$: 28×28 grey values in $[0, 1]$, label: the digit



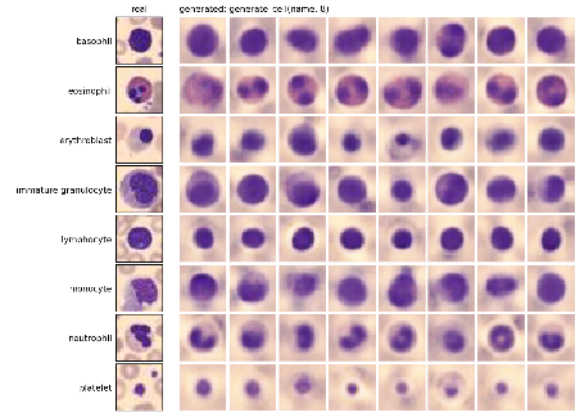
BloodMNIST (Lecture 4): 17 092 microscope images of single blood cells from healthy donors; $\mathbf{x}_i$: 28×28 colour pixels, label: one of 8 cell types, assigned by clinical pathologists

Y. LeCun et al., Proc. IEEE **86**, 2278 (1998), doi:10.1109/5.726791; A. Acevedo et al., Data in Brief **30**, 105474 (2020), doi:10.1016/j.dib.2020.105474; J. Yang et al., Sci. Data **10**, 41 (2023), doi:10.1038/s41597-022-01721-8

Generated Digits and Blood Cells



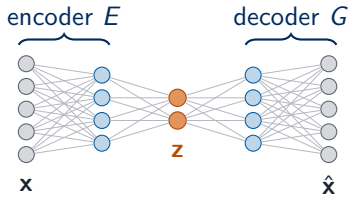
new digits: no image of the training set

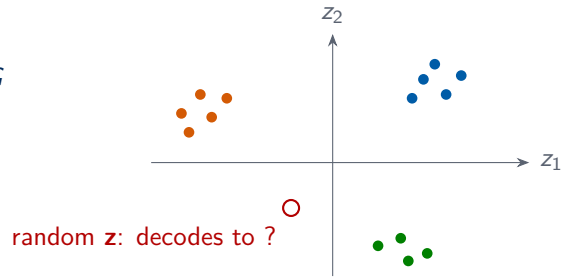


blood cells of a chosen type (first column: a real cell)

Both made by the models of this lecture, trained on a laptop CPU in minutes.

Autoencoder and Its Latent Space

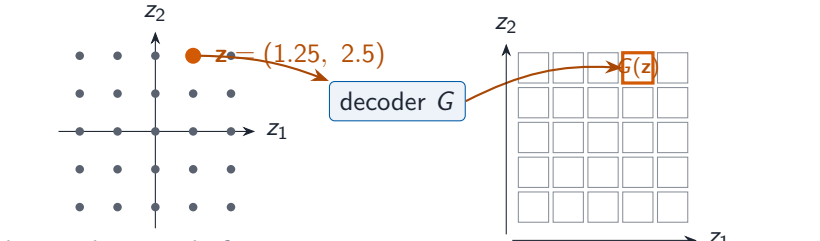




codes $E(\mathbf{x}_i)$ of training images (colour: digit)

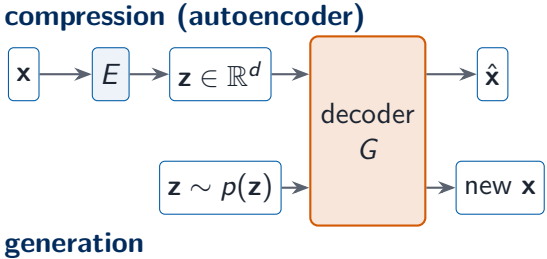
- trained only to rebuild its inputs: each image becomes one point $\mathbf{z}_i = E(\mathbf{x}_i)$
- nothing fixes where the points lie: no known distribution to draw a new $\mathbf{z}$ from
- the decoder never saw the gaps: a $\mathbf{z}$ there decodes to something that is no digit

Latent Grid



- a regular grid of latent points $\mathbf{z} = (z_1, z_2)$, $z_1, z_2 \in [-2.5, 2.5]$; each is decoded and its image placed at the same position (column: z_1, row: z_2)
- the sheet is a map of what the decoder produces everywhere in the latent plane, also where no training image was mapped

Compression and Generative Modelling



store each $\mathbf{x}_i$ in d numbers:

$$\min_{E,G} \frac{1}{N} \sum_{i=1}^{N} \|\mathbf{x}_i - G(E(\mathbf{x}_i))\|^2$$

new $\mathbf{x}$ distributed like the data:

$$p_\theta(\mathbf{x}) = \int p_\theta(\mathbf{x} \mid \mathbf{z})\, p(\mathbf{z})\, d\mathbf{z} \approx p_{\text{data}}(\mathbf{x})$$

E: encoder; N: number of training images $\mathbf{x}_i$; θ: weights of the decoder; $p_\theta(\mathbf{x} \mid \mathbf{z})$: the probability it assigns to $\mathbf{x}$ given $\mathbf{z}$; p_{data}: the distribution of the data.

- generation reuses the decoder of compression; missing are (i) a known $p(\mathbf{z})$ to draw from, (ii) a decoder trained for every $\mathbf{z}$ that $p(\mathbf{z})$ produces, not only the N codes
- **the VAE adds both:** each code becomes a Gaussian cloud; a KL term (a distance between distributions) pulls the clouds towards $\mathcal{N}(0, I)$

Compression and Generation Compared

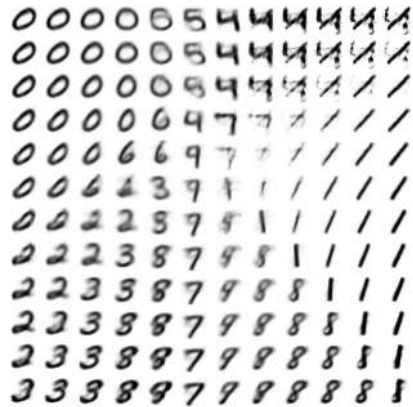
	compression (autoencoder)	generation
input	an image $\mathbf{x}$	a random $\mathbf{z} \sim p(\mathbf{z})$
output	$\hat{\mathbf{x}} = G(E(\mathbf{x})) \approx \mathbf{x}$	a new $\mathbf{x}$, not in the training set
objective	reconstruction error $\frac{1}{N}\sum_i \|\mathbf{x}_i - G(E(\mathbf{x}_i))\|^2$	$p_\theta(\mathbf{x}) \approx p_{\text{data}}(\mathbf{x})$
learned	encoder E and decoder G	decoder G; $p(\mathbf{z})$ chosen, $\mathcal{N}(0, I)$
latent codes	N points $\mathbf{z}_i = E(\mathbf{x}_i)$	a density $p(\mathbf{z})$ over a whole region
use	storage, denoising, features for other tasks	new samples: data augmentation, simulation

An Autoencoder Cannot Generate

Can the decoder of an autoencoder produce *new* digits? Pick a latent point $\mathbf{z}$ and compute $G(\mathbf{z})$.

- the autoencoder only learned to rebuild its training images; nothing tells us where in latent space the valid $\mathbf{z}$ are
- between and around the training codes there are regions that decode to images that are no digit

The sheet shows what the decoder does where it was never trained.



autoencoder with $d = 2$: images decoded from a regular grid of latent points

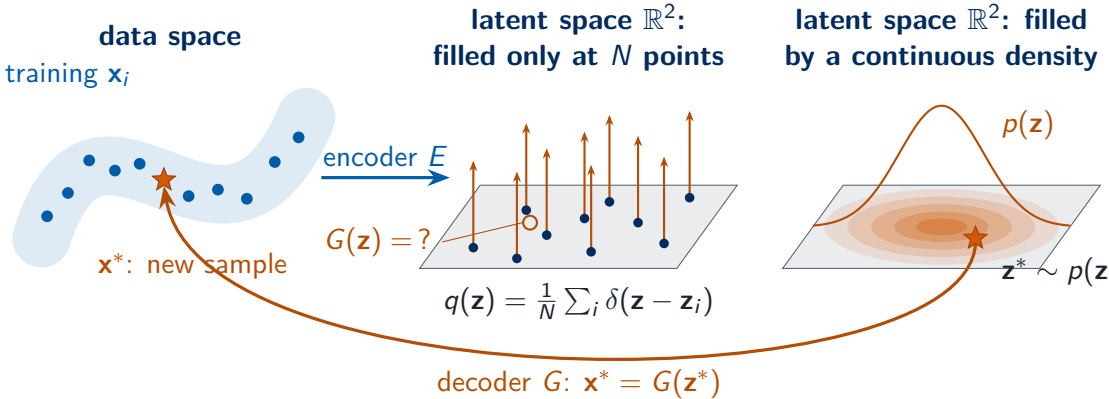
A Finite Set of Latent Codes

1. the latent space $\mathbb{R}^d$ (for $d = 2$ the plane (z_1, z_2)) is continuous: every $\mathbf{z} \in \mathbb{R}^d$ is a valid input of the decoder, $G(\mathbf{z})$ is defined
2. the autoencoder fills it only at N points, one per training image: $\mathbf{z}_i = E(\mathbf{x}_i)$, $i = 1, \ldots, N$ (MNIST: $N = 60\,000$ coloured dots)
3. the decoder was trained only at these points, so the only latent distribution the autoencoder provides is a sum of spikes,

$$q(\mathbf{z}) = \frac{1}{N} \sum_{i=1}^{N} \delta(\mathbf{z} - \mathbf{z}_i), \qquad \delta : \text{Dirac delta, a spike at } \mathbf{0} \text{ with } \int \delta(\mathbf{z})\, d\mathbf{z} = 1$$

   a draw from q returns some $\mathbf{z}_i$, and $G(\mathbf{z}_i) \approx \mathbf{x}_i$ rebuilds a training image: nothing new. For any other $\mathbf{z}$, $G(\mathbf{z})$ is unconstrained
4. wanted instead: a continuous density $p(\mathbf{z}) > 0$ over a whole region, with the decoder trained on all of it; the VAE turns each dot into a Gaussian cloud

Discrete Codes and a Continuous Latent Density



Middle: one spike per code $\mathbf{z}_i = E(\mathbf{x}_i)$, untrained gaps between them. Right: a draw from a density decodes to a new sample between the training images.

Generation with a Latent Space

The goal: new instances $\mathbf{x}$, drawn from (an approximation of) the distribution $p(\mathbf{x})$ of the data. The recipe of a latent-variable model:

$$\underbrace{\mathbf{z} \sim p(\mathbf{z})}_{\text{a random latent point}} \quad \longrightarrow \quad \underbrace{\mathbf{x} = G(\mathbf{z})}_{\text{decode it}}$$

It works only if two things hold:

1. $p(\mathbf{z})$ is *known* and easy to sample: we choose $p(\mathbf{z}) = \mathcal{N}(0, I)$
2. *every* $\mathbf{z}$ that $p(\mathbf{z})$ produces decodes to a plausible $\mathbf{x}$: the training data must fill the region where $p(\mathbf{z})$ puts its probability, without gaps

A finite set of codes satisfies neither. The variational autoencoder is built to satisfy both; its decoder becomes this generator G, and the model of the data is $p_\theta(\mathbf{x}) = \int p_\theta(\mathbf{x} \mid \mathbf{z})\, p(\mathbf{z})\, d\mathbf{z}$, which should approximate $p_{\text{data}}(\mathbf{x})$.

2
Probability



JAGIELLONIAN UNIVERSITY
IN KRAKOW

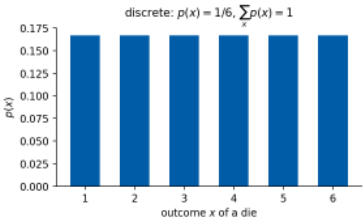
Discrete and Continuous Distributions

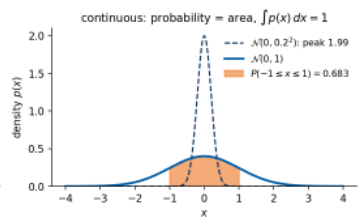
Discrete x (a die, a class, a spin): $p(x)$ is a probability,

$$0 \le p(x) \le 1, \qquad \sum_x p(x) = 1.$$

Continuous x (an intensity, a coordinate): $p(x)$ is a *density*; only areas are probabilities,

$$P(a \le x \le b) = \int_a^b p(x)\,dx, \qquad \int p(x)\,dx = 1$$



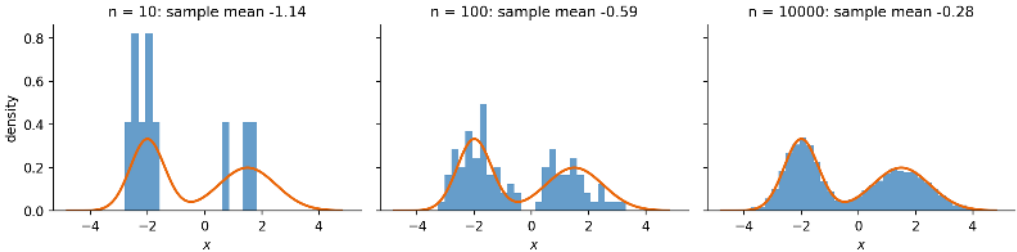


A density can exceed 1 (peak 1.99 for $\sigma = 0.2$). An MNIST image is a point $\mathbf{x} \in \mathbb{R}^{784}$, and $p(\mathbf{x})$ is a density in 784 dimensions.

Expectation and Averages over Samples

The **expectation** of a function f under p, and its estimate from samples $x_1, \ldots, x_n$ of p:

$$\mathbb{E}_{x\sim p}[f(x)] = \int p(x)\, f(x)\, dx \;\approx\; \frac{1}{n}\sum_{i=1}^{n} f(x_i), \qquad \text{std of the estimate} = \frac{\sigma_f}{\sqrt{n}}$$



Orange: $p(x) = \frac{1}{2}\mathcal{N}(x; -2, 0.6^2) + \frac{1}{2}\mathcal{N}(x; 1.5, 1)$, with $\mathbb{E}[x] = -0.25$; blue: histograms of n samples. Every loss in this lecture is an expectation, estimated as an average over a mini-batch (σ_f: standard deviation of $f(x)$ under p).

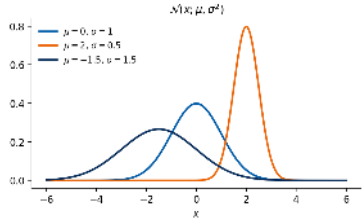
The Gaussian Distribution

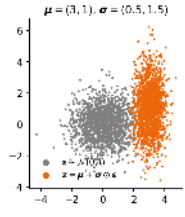
One dimension, mean μ, standard deviation σ:

$$\mathcal{N}(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/2\sigma^2}$$

d independent coordinates; $\mathcal{N}(0, I)$: all $\mu_k = 0$, $\sigma_k = 1$:

$$\mathcal{N}(\mathbf{z}; \boldsymbol{\mu}, \operatorname{diag} \boldsymbol{\sigma}^2) = \prod_{k=1}^{d} \mathcal{N}(z_k; \mu_k, \sigma_k^2)$$



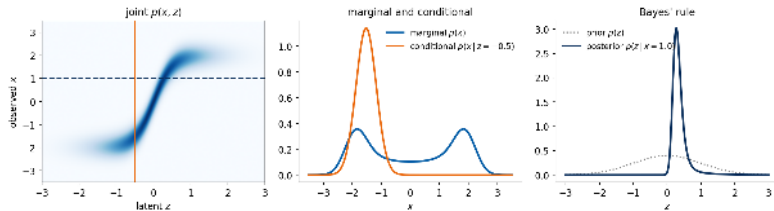


Sampling: draw $\varepsilon \sim \mathcal{N}(0, I)$, then $\mathbf{z} = \boldsymbol{\mu} + \boldsymbol{\sigma} \odot \varepsilon$ ($\odot$: element-wise product) is a sample of $\mathcal{N}(\boldsymbol{\mu}, \operatorname{diag} \boldsymbol{\sigma}^2)$: a shift and a stretch.

Joint, Conditional and Marginal Distributions

An observed x and a hidden (latent) z, with joint distribution $p(x, z)$:

$$\underbrace{p(x) = \int p(x, z)\, dz}_{\text{marginal}} \qquad \underbrace{p(x, z) = p(x \mid z)\, p(z)}_{\text{conditional, product rule}} \qquad \underbrace{p(z \mid x) = \frac{p(x \mid z)\, p(z)}{p(x)}}_{\text{Bayes' rule: posterior}}$$



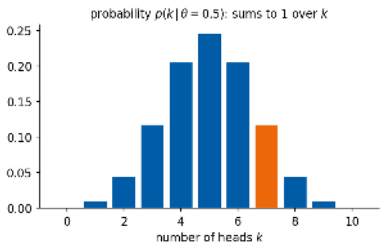
$p(z)$: the *prior*, what we know about z before seeing x; $p(z \mid x)$: the *posterior*, after seeing x. Here $z \sim \mathcal{N}(0, 1)$, $x \mid z \sim \mathcal{N}\big(2 \tanh(2z), 0.35^2\big)$: a two-peaked $p(x)$; observing $x = 1$ narrows the prior (std 1) to a posterior with std 0.25.

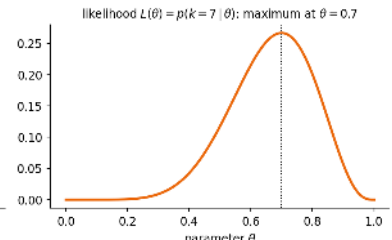
Probability and Likelihood

A model $p(x \mid \theta)$: the probability of data x when the parameters are θ. One formula, read in two ways:

probability: θ fixed, x varies; summed over all outcomes x it gives 1

likelihood $L(\theta) = p(x_{\text{obs}} \mid \theta)$: data fixed, θ varies; not a distribution over θ





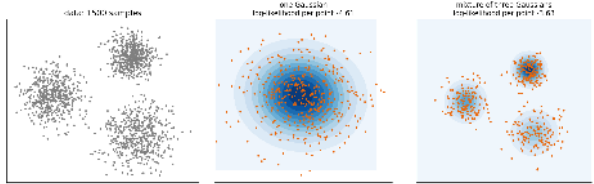
A coin with probability θ of heads, tossed 10 times: $p(k \mid \theta) = \binom{10}{k}\theta^k(1-\theta)^{10-k}$; $p(7 \mid 0.5) = 0.117$. With 7 heads observed, $L(\theta)$ peaks at $\theta = 0.7$, and $\int_0^1 L\, d\theta = 1/11$, not 1.

Maximum Likelihood

Independent samples $\mathbf{x}_1, \ldots, \mathbf{x}_n$; a model $p_\theta(\mathbf{x})$. The likelihood of all the data, and its logarithm:

$$L(\theta) = \prod_{i=1}^{n} p_\theta(\mathbf{x}_i), \qquad \ell(\theta) = \sum_{i=1}^{n} \log p_\theta(\mathbf{x}_i), \qquad \theta^* = \arg\max_\theta \ell(\theta)$$

The logarithm turns the product into a sum and does not move the maximum. One Gaussian: θ^* is the sample mean and variance (with $1/n$).



1500 points from three groups; blue: fitted density, orange: 400 new samples. Mean log-likelihood per point: -4.61 (one Gaussian), -3.63 (mixture of three).

The Kullback–Leibler Divergence

Mismatch of a model q with the true p:

$$\mathrm{KL}(p \,\|\, q) = \mathbb{E}_{x\sim p}\Big[\log \frac{p(x)}{q(x)}\Big]$$

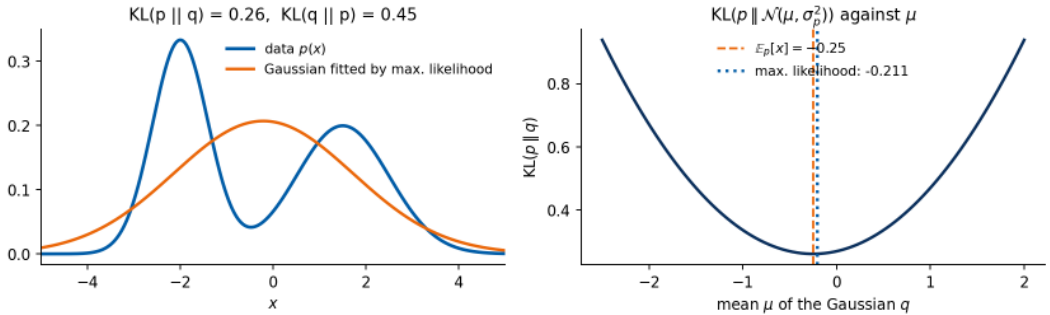
≥ 0, $= 0$ only if $q = p$, not symmetric (so not a distance). With $-\log q(x)$ the surprise of seeing x under q: the average extra surprise when data from p are described by q.

With $q = p_\theta$, split the logarithm:

$$\mathrm{KL}(p \,\|\, p_\theta) = \underbrace{\mathbb{E}_p[\log p]}_{\text{no } \theta} - \underbrace{\mathbb{E}_p[\log p_\theta]}_{\approx \ell(\theta)/n}$$

maximum likelihood = minimum $\mathrm{KL}(p \,\|\, p_\theta)$

Maximum Likelihood and the KL Divergence

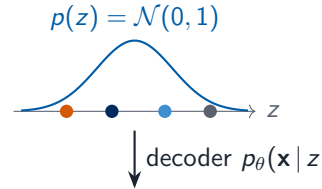


One Gaussian (width fixed at σ_p, the std of p) fitted to the two-peaked p: the KL is smallest at $\mu = \mathbb{E}_p[x] = -0.25$; maximum likelihood on 10^4 samples gives -0.21.

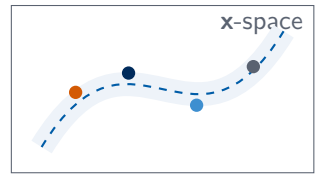
Latent-Variable Models

A simple hidden $\mathbf{z}$ and a map to the data:

$$p_\theta(\mathbf{x}) = \int p_\theta(\mathbf{x} \mid \mathbf{z})\, p(\mathbf{z})\, d\mathbf{z}$$



- **generate**: draw $\mathbf{z} \sim p(\mathbf{z})$, then $\mathbf{x} \sim p_\theta(\mathbf{x} \mid \mathbf{z})$
- Gaussian mixture: $z \in \{1, \ldots, K\}$ with $p(z = k) = \pi_k$, $p(\mathbf{x} \mid z = k) = \mathcal{N}(\boldsymbol{\mu}_k, \Sigma_k)$
- VAE: $\mathbf{z} \in \mathbb{R}^d$, $p(\mathbf{z}) = \mathcal{N}(0, I)$, and a neural network (the decoder) sets $p_\theta(\mathbf{x} \mid \mathbf{z})$



each z gives a small cloud around a point of a curve in data space

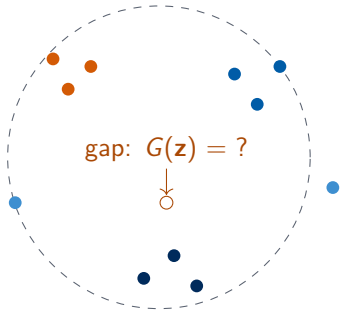
3

Variational Autoencoders

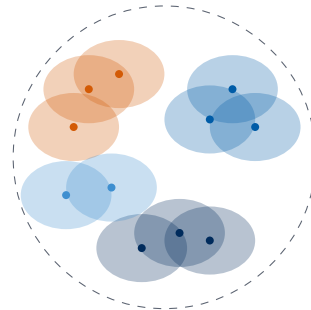
JAGIELLONIAN UNIVERSITY
IN KRAKÓW

Latent Points and Latent Clouds

autoencoder: one point per input



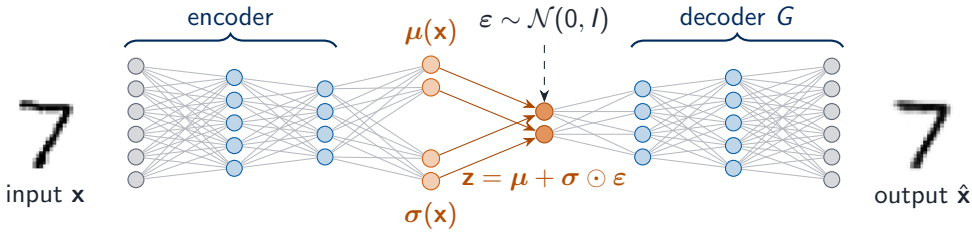
VAE: a Gaussian cloud per input



dashed circle: where $\mathcal{N}(0, I)$ puts most of its probability

Colours: four kinds of input. For each input $\mathbf{x}$ the VAE encoder outputs a centre $\boldsymbol{\mu}(\mathbf{x})$ and widths $\boldsymbol{\sigma}(\mathbf{x})$: the cloud $\mathcal{N}(\boldsymbol{\mu}(\mathbf{x}), \operatorname{diag} \boldsymbol{\sigma}^2(\mathbf{x}))$. Neighbouring clouds overlap, and together they cover the dashed circle.

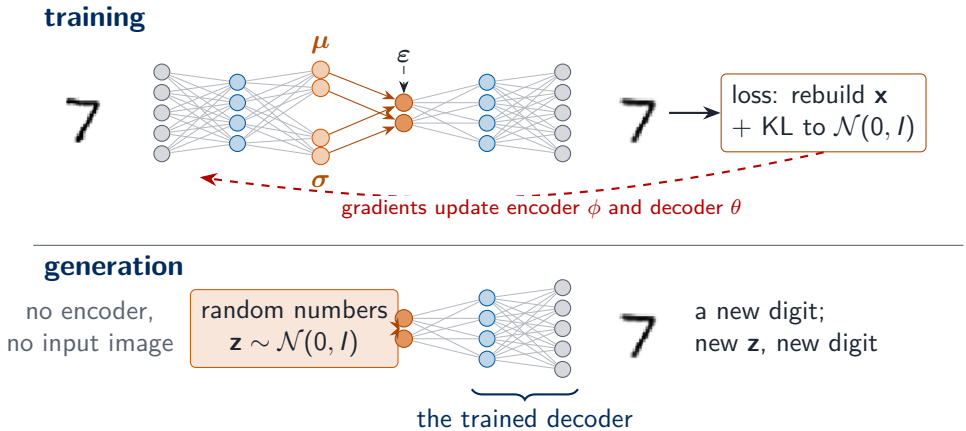
Variational Autoencoder



- an autoencoder maps an input to one *point* $\mathbf{z}$; a VAE maps it to a Gaussian *cloud* centred at $\mu(\mathbf{x})$ with widths $\sigma(\mathbf{x})$ (one μ_k, σ_k for each of the d latent coordinates)
- during training a random point of the cloud is decoded, so every point near $\mu(\mathbf{x})$ must decode to something like $\mathbf{x}$
- to *generate*, skip the encoder: draw $\mathbf{z} \sim \mathcal{N}(0, I)$ and compute $G(\mathbf{z})$

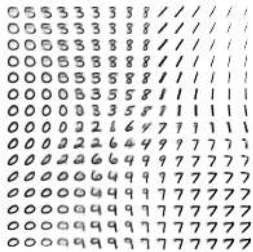
D. P. Kingma, M. Welling, *Auto-Encoding Variational Bayes*, ICLR 2014, arXiv:1312.6114

Training and Sampling a VAE



The KL term has placed the training clouds inside $\mathcal{N}(0, I)$, so random numbers drawn from it land where the decoder has learned to produce digits.

New Digits from the VAE



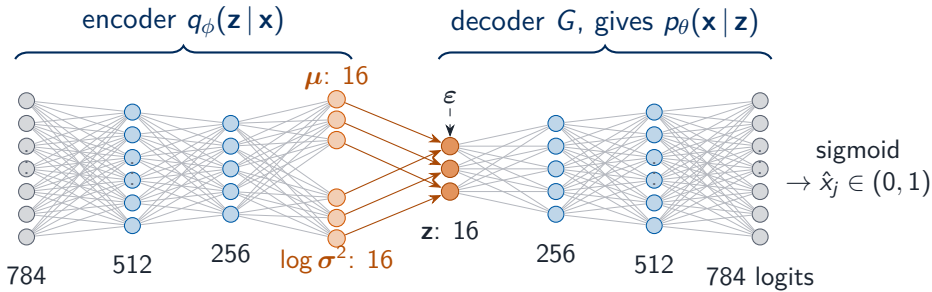
$d = 2$: images decoded from a grid of $\mathbf{z}$ over $[-2.5, 2.5]^2$



$d = 16$: 32 new digits, each from a random $\mathbf{z} \sim \mathcal{N}(0, I)$

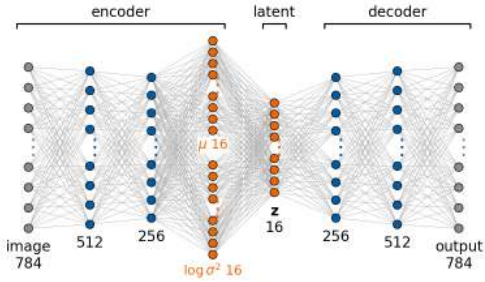
- continuous latent space: neighbouring $\mathbf{z}$ give similar digits
- new, not memorised: on average a generated digit is as far from its nearest training image (Euclidean distance in pixel space, 4.73) as a real test digit the network never saw (4.03); somewhat blurry, a known property of VAEs

The VAE on MNIST: Layers



Dense layers with ReLU (a few nodes drawn per layer; the numbers give the true sizes); logits: the outputs before the sigmoid. ϕ: weights of the encoder, θ: of the decoder. The encoder outputs $\log \sigma^2$, which can take any sign; $d = 2$ or 16. Trained 20 epochs with Adam on the 60 000 training digits, without labels.

VAE Layers and Parameters



layer	output	param.
encoder, input $\mathbf{x}$*: 784*		
dense + ReLU	512	401 920
dense + ReLU	256	131 328
dense $\to \boldsymbol{\mu}$	16	4 112
dense $\to \log \boldsymbol{\sigma}^2$	16	4 112
decoder, input $\mathbf{z}$*: 16*		
dense + ReLU	256	4 352
dense + ReLU	512	131 584
dense (logits)	784	402 192

$d = 16$ latent coordinates; a dense layer from n to m nodes has $nm + m$ parameters. Total 1 079 600 parameters, all trainable (torchinfo).

The Decoder Defines $p_\theta(\mathbf{x} \mid \mathbf{z})$

The decoder ends with a sigmoid, $\hat{\mathbf{x}} = G(\mathbf{z}) \in (0,1)^{784}$ (its last layer outputs *logits*, the numbers before the sigmoid); $\hat{x}_j$ is the probability that pixel j is ink. For black/white pixels $x_j \in \{0,1\}$, drawn independently given $\mathbf{z}$:

$$p_\theta(\mathbf{x} \mid \mathbf{z}) = \prod_{j=1}^{784} \hat{x}_j^{\,x_j} \, (1-\hat{x}_j)^{1-x_j} \quad \Longrightarrow \quad -\log p_\theta(\mathbf{x} \mid \mathbf{z}) = -\sum_{j=1}^{784} \Big[x_j \log \hat{x}_j + (1-x_j) \log(1-\hat{x}_j) \Big]$$

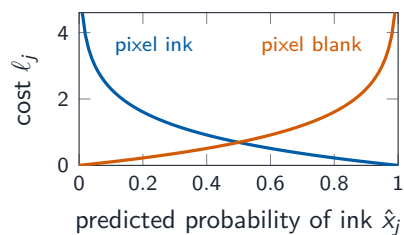
the reconstruction loss (binary cross-entropy) is minus a log-likelihood. MNIST pixels are grey, $x_j \in [0,1]$: the same formula is still used as the loss.

Binary Cross-Entropy

Pixel j is ink ($x_j = 1$) or blank ($x_j = 0$); the decoder gives the probability of ink $\hat{x}_j$. The cost of pixel j is minus the log-probability given to what it actually is; the reconstruction loss is $\sum_{j=1}^{784} \ell_j$:

$$\ell_j = -\big[x_j \log \hat{x}_j + (1 - x_j) \log(1 - \hat{x}_j)\big] = \begin{cases} -\log \hat{x}_j, & x_j = 1 \text{ (ink)} \\ -\log(1 - \hat{x}_j), & x_j = 0 \text{ (blank)} \end{cases}$$

- $\ell_j = 0$ only if the decoder is certain and right; it grows without bound if it is confident and wrong
- *binary*: two outcomes per pixel; *cross-entropy*: the surprise $-\log$ of the true outcome (classification loss, two classes)



Gaussian Pixels and Independent Pixels

Gaussian pixels give the squared error

$p_\theta(\mathbf{x} \mid \mathbf{z}) = \mathcal{N}(\mathbf{x}; G(\mathbf{z}), \sigma_x^2 I)$:

$$-\log p_\theta = \frac{\|\mathbf{x} - G(\mathbf{z})\|^2}{2\sigma_x^2} + \text{const}$$

Independent pixels

given $\mathbf{z}$, the product treats pixels as unrelated: all correlations between pixels must come from $\mathbf{z}$.

The Evidence Lower Bound

Maximum likelihood needs $\log p_\theta(\mathbf{x}) = \log \int p_\theta(\mathbf{x} \mid \mathbf{z})\, p(\mathbf{z})\, d\mathbf{z}$: an integral over all $\mathbf{z}$, out of reach. The encoder $q_\phi(\mathbf{z} \mid \mathbf{x})$ approximates the posterior $p_\theta(\mathbf{z} \mid \mathbf{x})$. Since $p_\theta(\mathbf{x}) = p_\theta(\mathbf{x}, \mathbf{z}) / p_\theta(\mathbf{z} \mid \mathbf{x})$ for every $\mathbf{z}$, average log of it over $\mathbf{z} \sim q_\phi$ and multiply inside by q_ϕ / q_ϕ:

$$\log p_\theta(\mathbf{x}) = \mathbb{E}_{q_\phi}\Big[\log \frac{p_\theta(\mathbf{x}, \mathbf{z})}{q_\phi(\mathbf{z} \mid \mathbf{x})}\Big] + \mathbb{E}_{q_\phi}\Big[\log \frac{q_\phi(\mathbf{z} \mid \mathbf{x})}{p_\theta(\mathbf{z} \mid \mathbf{x})}\Big]$$

With $p_\theta(\mathbf{x}, \mathbf{z}) = p_\theta(\mathbf{x} \mid \mathbf{z})\, p(\mathbf{z})$:

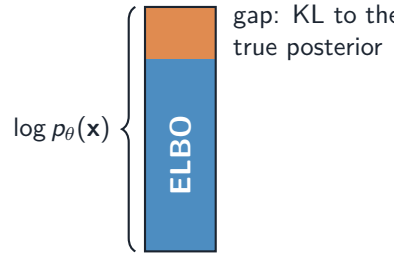
$$\log p_\theta(\mathbf{x}) = \underbrace{\mathbb{E}_{q_\phi}\big[\log p_\theta(\mathbf{x} \mid \mathbf{z})\big] - \mathrm{KL}(q_\phi \,\|\, p(\mathbf{z}))}_{\text{ELBO}} + \underbrace{\mathrm{KL}(q_\phi \,\|\, p_\theta(\mathbf{z} \mid \mathbf{x}))}_{\geq 0}$$

The VAE Loss

Per input $\mathbf{x}$, with one sample $\mathbf{z} = \boldsymbol{\mu} + \boldsymbol{\sigma} \odot \boldsymbol{\varepsilon}$:

$$\mathcal{L}(\mathbf{x}) = \underbrace{-\log p_\theta(\mathbf{x} \mid \mathbf{z})}_{\text{reconstruction (BCE)}} + \underbrace{\tfrac{1}{2} \sum_{k=1}^{d} \left(\mu_k^2 + \sigma_k^2 - \ln \sigma_k^2 - 1\right)}_{\mathrm{KL}(\mathcal{N}(\boldsymbol{\mu}, \mathrm{diag}\, \boldsymbol{\sigma}^2) \,\|\, \mathcal{N}(0, I))}$$

- $\mathcal{L} = -\text{ELBO}$; since $\mathrm{KL} \geq 0$, $-\mathcal{L} \leq \log p_\theta(\mathbf{x})$: minimising $\mathcal{L}$ raises a lower bound on the log-likelihood
- each KL term is 0 only at $\mu_k = 0$, $\sigma_k = 1$
- the two terms pull apart: reconstruction wants small clouds far from each other, the KL pulls every cloud to $\mathcal{N}(0, I)$; the balance: overlapping clouds that fill $\mathcal{N}(0, I)$ without gaps



$d = 16$ VAE, mean over the training digits: $\mathcal{L} = 99.9 = 80.3$ (reconstruction) $+ 19.6$ (KL)

The Reparametrisation Trick

Training needs $\partial\mathcal{L}/\partial\boldsymbol{\mu}$ and $\partial\mathcal{L}/\partial\boldsymbol{\sigma}$, but $\mathbf{z}$ is *drawn at random*: "draw $\mathbf{z}$ from $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\sigma}^2)$" has no derivative with respect to $\boldsymbol{\mu}$ and $\boldsymbol{\sigma}$.

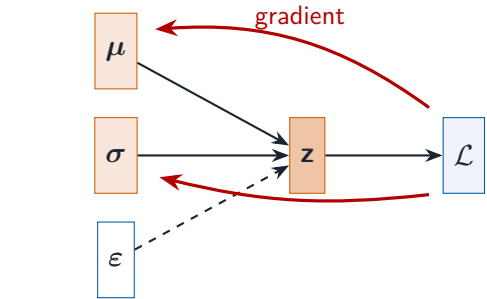
Move the randomness out of the network:

$$\mathbf{z} = \boldsymbol{\mu} + \boldsymbol{\sigma} \odot \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, I)$$

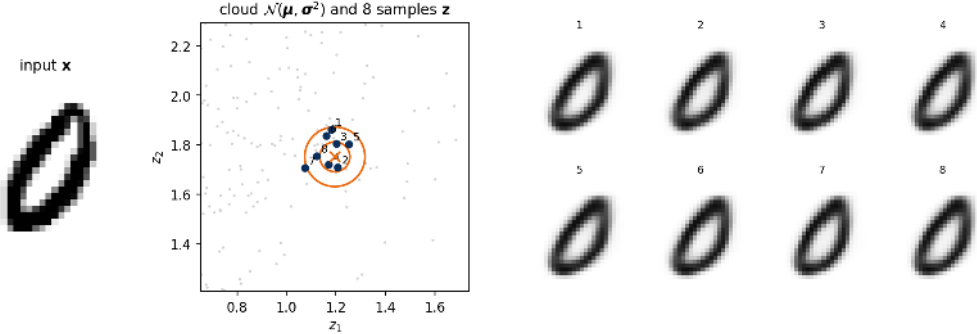
ε is an input that does not depend on the weights; $\mathbf{z}$ is an ordinary function of $\boldsymbol{\mu}$ and $\boldsymbol{\sigma}$:

$$\frac{\partial z_k}{\partial \mu_k} = 1, \qquad \frac{\partial z_k}{\partial \sigma_k} = \varepsilon_k,$$

and backpropagation runs from the loss through the decoder, $\mathbf{z}$ and the encoder.



Samples from the Cloud of One Input



One test digit (the 0 that the $d = 2$ VAE rebuilds best; with two latent numbers some unusual digits are rebuilt as another digit). The encoder gives its cloud: centre $\boldsymbol{\mu}$ (cross), widths $\boldsymbol{\sigma}$ (ellipses: 1 and 2σ). Eight noises $\boldsymbol{\varepsilon} \sim \mathcal{N}(0, I)$ give $\mathbf{z} = \boldsymbol{\mu} + \boldsymbol{\sigma} \odot \boldsymbol{\varepsilon}$ (dots); the decoder turns each into a version of the input. In training, one such $\mathbf{z}$ per input is decoded and compared with $\mathbf{x}$; autograd gives $\partial z_1/\partial \mu_1 = 1$, $\partial z_1/\partial \sigma_1 = \varepsilon_1$.

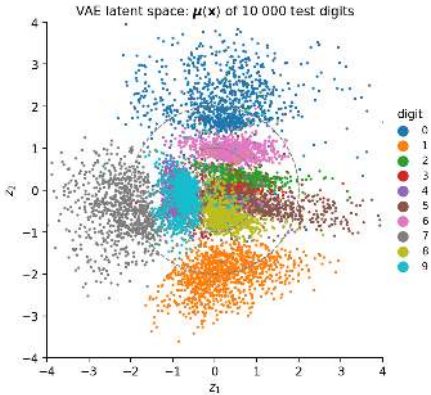
Sampling a VAE

One draw is two random choices, *ancestral sampling* of the model $p_\theta(\mathbf{x}) = \int p_\theta(\mathbf{x} \mid \mathbf{z})\, p(\mathbf{z})\, d\mathbf{z}$:

$$\mathbf{z} \sim \mathcal{N}(0, I), \qquad \mathbf{x} \sim p_\theta(\mathbf{x} \mid \mathbf{z}) \;\; \text{(each pixel ink with probability } \hat{x}_j(\mathbf{z})\text{)}$$

- in practice we show the *mean image* $\hat{\mathbf{x}}(\mathbf{z}) = G(\mathbf{z})$, the pixel probabilities, rather than one random draw of the pixels: the grey levels are probabilities of ink
- $\mathbf{z}$ carries the content (which digit, its slant and thickness); the pixel draw only adds independent noise
- **reconstruction** (encode $\mathbf{x}$, decode) reproduces a given image; **generation** (draw $\mathbf{z}$, decode) makes a new one
- **interpolation**: encode two digits to $\mathbf{z}_A$, $\mathbf{z}_B$ and decode $(1-t)\,\mathbf{z}_A + t\,\mathbf{z}_B$: one digit turns smoothly into the other, because the latent space is filled without gaps

The Latent Space of a VAE

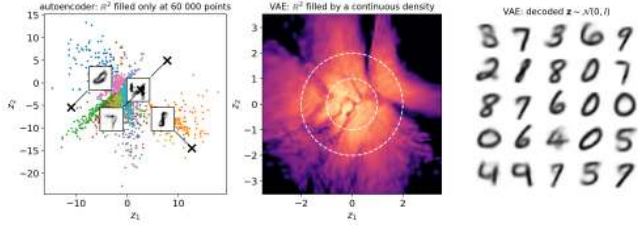


Latent representation: the encoder compresses $\mathbf{x} \in \mathbb{R}^{784}$ into $\mathbf{z} \in \mathbb{R}^d$, coordinates in which similar images are close.

VAE with $d = 2$, trained without labels; each point: the centre $\boldsymbol{\mu}(\mathbf{x})$ of one test digit's cloud, coloured by its label.

- digits of one class gather: the latent space organises the data
- 80% of the centres lie within radius 2 (dashed: radius 1, 2), against 86% for $\mathcal{N}(0, I)$; classes 0, 1, 7 sit at the edge

Codes of the Trained Autoencoder and VAE



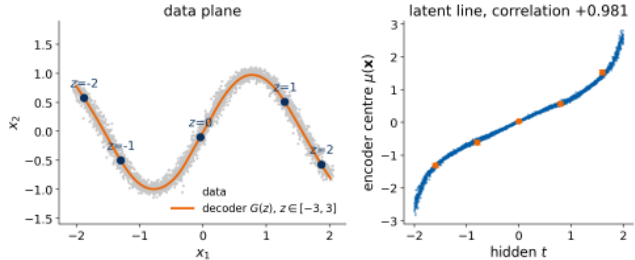
- autoencoder ($d = 2$): 60 000 training codes spread to $|z| \approx 20$; latent points in the gaps ($\times$) decode to images that are no digit
- VAE: the median cloud width is 5.6 times the median spacing of neighbouring codes, so the clouds overlap into a continuous density (dashed: 1 and 2 standard deviations of $\mathcal{N}(0, I)$)
- only 3.6 % of draws $\mathbf{z} \sim \mathcal{N}(0, I)$ land where the density of the clouds is below 10 % of the prior; decoded draws are digits

4

Latent Spaces with Known Causes

JAGIELLONIAN UNIVERSITY
IN KRAKOW

A Curve with One Hidden Number



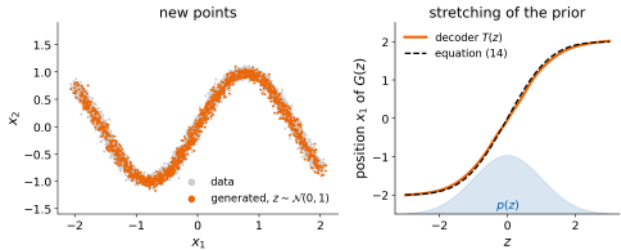
Data: $N = 4000$ points

$$\mathbf{x} = (t,\ \sin 2t) + \text{noise}, \quad t \in [-2, 2]$$

uniform t, Gaussian noise of width 0.05.
One hidden number, two data coordinates.

- VAE with $d = 1$, Gaussian decoder of width 0.05
- the encoder centre μ orders the points like t: rank correlation 0.9996 (linear 0.981: an S-shaped relation)
- clouds narrow: median $\sigma = 0.028$

Generated Points and the Stretching of the Prior



Decoder position $T(z)$ along the curve; a latent interval dz is spread over $|dT/dz|\,dz$, so generated points have density $p(t) = p(z)/|dT/dz|$. A uniform t on $[-2, 2]$ requires

$$T(z) = 4\,\Phi(z) - 2, \qquad \Phi(z) = \int_{-\infty}^{z} \mathcal{N}(z'; 0, 1)\,dz'$$

the inverse of the Gaussian cumulative distribution. The decoder found it by itself: deviation at most 0.12 for $|z| \le 2.5$; 99.2 % of generated points lie in $[-2, 2]$.

The KL Weight on the Curve

β-weighted loss: reconstruction $+ \beta \cdot \mathrm{KL}\big(q_\phi(z|\mathbf{x}) \,\|\, p(z)\big)$

β	median σ	range of μ	correlation of μ with t
0.02	0.0065	-3.99 to 4.04	0.989
1	0.028	-2.83 to 2.81	0.981
20	0.13	-2.20 to 2.19	0.985

- β small: a sharp code, but the centres spread beyond the prior; samples from $\mathcal{N}(0, 1)$ miss the ends of the curve, the autoencoder problem returns
- β large: wide clouds; one output for a whole stretch of the curve is its average, so the decoder cuts the corners of the sine
- the *order* of the latent values stays the same; β changes how finely they resolve t and how well they fill the prior

A Continuous Mixture Model

The VAE is a mixture with a continuous label z:

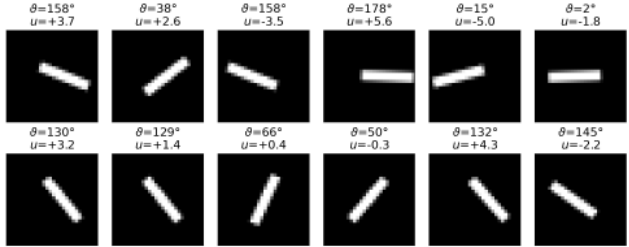
$$p_\theta(\mathbf{x}) = \int p_\theta(\mathbf{x} \mid z)\, p(z)\, dz$$

Mean log-likelihood of 2000 fresh points of the curve:

model	$\langle \log p(\mathbf{x}) \rangle$
mixture of 3 Gaussians	-0.850
mixture of 30 Gaussians	-0.388
VAE, $d = 1$ (integral over z computed numerically)	-0.401
true process (best possible)	-0.360

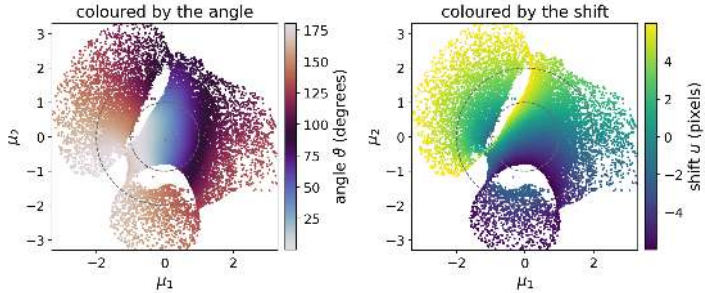
- the 30 component labels are an unordered list; the VAE's z is a coordinate along the curve, defined between the training points
- the ELBO, -0.552 on average, lies below $\log p$ for every point; the gap, 0.151 nats, is the KL divergence between the encoder clouds and the exact posteriors

Bars with an Angle and a Shift



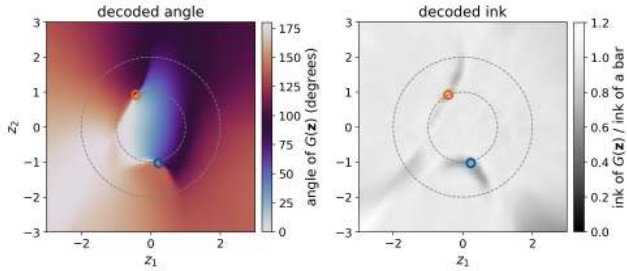
- 20 000 images of 28×28 pixels: one bar of length 16 and width 3 pixels, angle $\vartheta \in [0, \pi)$, horizontal shift $u \in [-6, 6]$ pixels
- 784 grey values fixed by two numbers: the images lie on a 2D surface in $\mathbb{R}^{784}$
- the angle is periodic, $\vartheta = \pi$ is $\vartheta = 0$: the surface is a *cylinder* (a circle of angles times an interval of shifts)
- VAE with $d = 2$, dense $784 \to 256 \to 128 \to 2$, Bernoulli decoder

The Latent Plane of the Bars



- the shift is read off mostly along μ_2: correlation $+0.82$ (μ_1: -0.36)
- the angle does *not* go round a clean ring: the polar angle of $\boldsymbol{\mu}$ follows 2ϑ only weakly (circular correlation 0.47)
- empty slits, and colour jumps across them: where the angle is not continuous

Vortices of the Decoded Angle



Angle of the decoded image $G(\mathbf{z})$ from its second moments (error 0.02° on training bars). Around a small loop, $2\hat{\vartheta}$ turns by a multiple of 2π: the *charge*.

- charge $+1$ near $(-0.42, 0.93)$, charge -1 near $(0.23, -1.02)$ (circles)
- seams start at the cores: faint or split images; latent points decoding to less than 85 % of the ink of a bar carry 5.5 % of the prior probability

Why the Plane Forces Vortices

The data surface is a cylinder; the latent space is a plane.

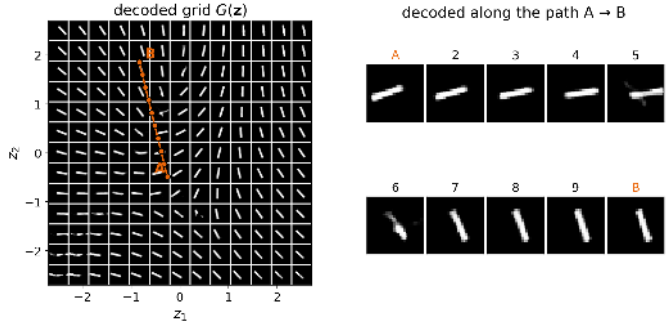
1. if every decoded image were a bar, $2\hat{\vartheta}(\mathbf{z})$ would be a continuous map from the plane to a circle; on a plane without holes such a map turns *zero* times around every closed loop
2. the bars of one shift form a closed loop of images, ϑ from 0 to π; the encoder maps them to a closed latent loop around which the decoded angle turns *once*
3. hence inside that loop the decoder must produce images without a defined angle: a vortex core, net charge $+1$

Arrangement of the Vortices

- topology fixes only the net charge; one vortex with bars on a ring, or a pair of opposite charges joined by seams, are both allowed
- this run: a $+1/-1$ pair near the origin; another random seed can arrange them differently
- the same mathematics as the vortices of the XY model of magnetism, where spin directions play the role of 2ϑ
- the Gaussian prior has no hole, so some prior probability always lands on images that are no proper bar

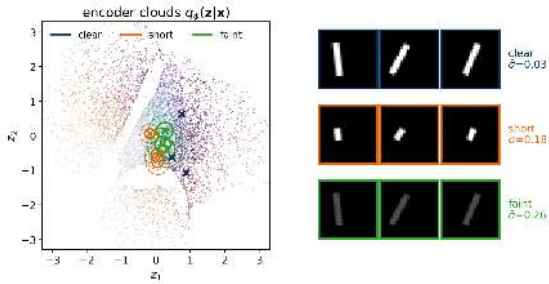
A latent space has a shape. When the data have a periodic cause, a Gaussian latent plane cannot represent them without defects.

A Straight Latent Path Crosses a Seam



From bar A ($\vartheta = 20°$, $u = -4$) to bar B ($\vartheta = 111°$, $u = +4$) on a straight latent line: images 5 and 6 are two crossed bars, because the line passes the $+1$ vortex. A path around the vortex would turn the bar smoothly: a straight latent line is not the natural route between two images.

Encoder Clouds and the Exact Posterior



With $d = 2$ the posterior $p_\theta(\mathbf{z}|\mathbf{x}) \propto p_\theta(\mathbf{x}|\mathbf{z})\, p(\mathbf{z})$ can be computed on a grid. Median widths:

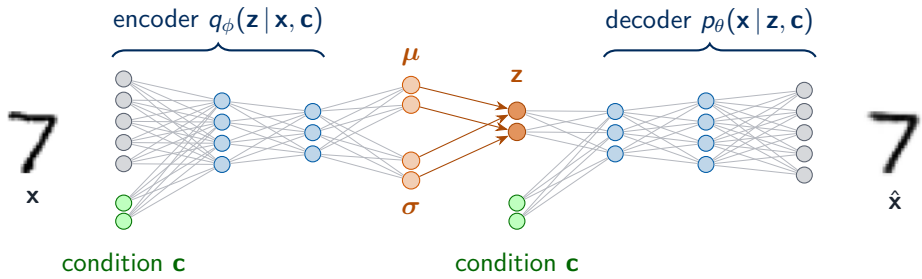
image	encoder σ	posterior
clear bar	0.027	0.026
short bar	0.183	0.026
faint bar	0.257	0.032
noisy bar	0.005	0.016

Clear bars: agreement. Short and faint bars (unseen in training): the encoder widens 7–10 times, the posterior hardly. Noisy bars: the encoder is overconfident. The encoder only *approximates* the posterior.

5
Conditional Generation

JAGIELLONIAN UNIVERSITY
IN KRAKOW

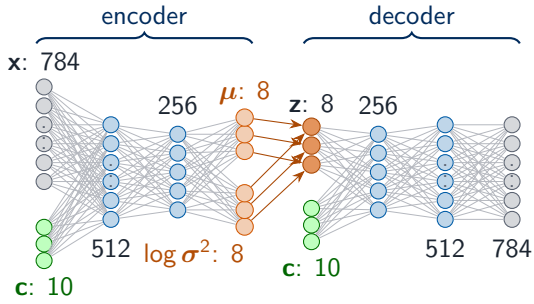
Conditional VAE



$\mathbf{c}$ is an extra input of *both* networks, $q_\phi(\mathbf{z} \mid \mathbf{x}, \mathbf{c})$ and $p_\theta(\mathbf{x} \mid \mathbf{z}, \mathbf{c})$; the model is $p_\theta(\mathbf{x} \mid \mathbf{c}) = \int p_\theta(\mathbf{x} \mid \mathbf{z}, \mathbf{c})\, p(\mathbf{z})\, d\mathbf{z}$, with the same loss as the VAE. To generate: fix $\mathbf{c}$, draw $\mathbf{z} \sim \mathcal{N}(0, I)$, compute $G(\mathbf{z}, \mathbf{c})$.

K. Sohn, H. Lee, X. Yan, *Learning Structured Output Representation using Deep Conditional Generative Models*, NeurIPS 2015, NeurIPS proceedings

Conditional VAE Layers and Parameters



layer	output	param.
encoder, input $[\mathbf{x}, \mathbf{c}]$: $784 + 10$		
dense + ReLU	512	407 040
dense + ReLU	256	131 328
dense $\to \boldsymbol{\mu}$	8	2 056
dense $\to \log \boldsymbol{\sigma}^2$	8	2 056
decoder, input $[\mathbf{z}, \mathbf{c}]$: $8 + 10$		
dense + ReLU	256	4 864
dense + ReLU	512	131 584
dense (logits)	784	402 192

$\mathbf{c}$: the one-hot digit, appended to the input of both networks; $d = 8$.
Total 1 081 120 parameters, all trainable (torchinfo).

Condition and Latent Vector

- $\mathbf{c}$ tells the decoder *what* to draw. Every bit stored in $\mathbf{z}$ costs KL, while what $\mathbf{c}$ supplies is free, so $\mathbf{z}$ keeps only what $\mathbf{c}$ does not say: the *style* (slant, thickness, size)

- a class k enters *one-hot*, $\mathbf{e}_k = (0, \ldots, 0, 1, 0, \ldots, 0)$ with the 1 at position k, not as the number k, which would impose an order on the classes; physical parameters such as (T, h) enter as numbers

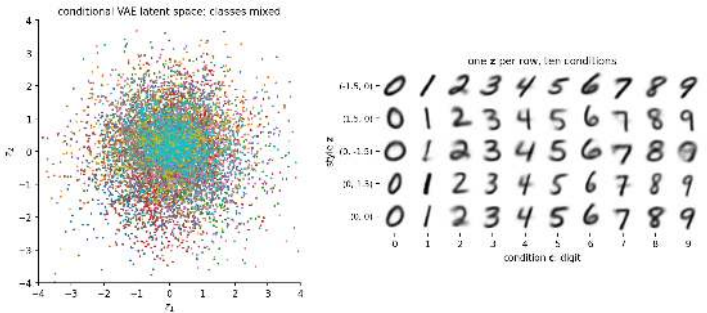
Sampling a Conditional VAE

The model is $p_\theta(\mathbf{x} \mid \mathbf{c}) = \int p_\theta(\mathbf{x} \mid \mathbf{z}, \mathbf{c})\, p(\mathbf{z})\, d\mathbf{z}$. Sampling: we *choose* $\mathbf{c}$ and *draw* $\mathbf{z}$:

$$\mathbf{c} \text{ given (what)}, \qquad \mathbf{z} \sim \mathcal{N}(0, I) \text{ (how)}, \qquad \hat{\mathbf{x}} = G(\mathbf{z}, \mathbf{c})$$

- one $\mathbf{c}$, many $\mathbf{z}$: many instances of the same class, different in style (digits 7 in many handwritings; cells of one type)
- one $\mathbf{z}$, many $\mathbf{c}$: the same style for every class (the rows of the digit grid on the next slides keep slant and thickness)
- continuous conditions work the same way: fix the temperature T and draw $\mathbf{z}$ to get spin configurations at that T
- the condition must be known for every training image: a conditional VAE needs labels; what the labels do not describe is left to $\mathbf{z}$

Latent Space of the Conditional VAE



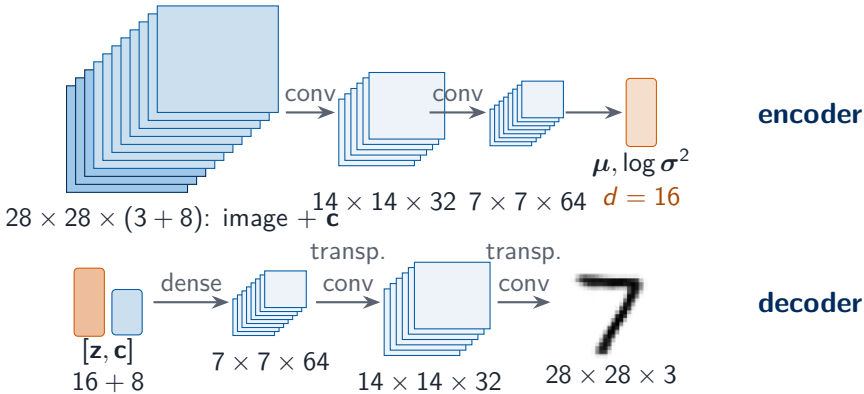
Conditional VAE with $d = 2$, label given as $\mathbf{c}$. Left: the centres $\boldsymbol{\mu}(\mathbf{x}, \mathbf{c})$ of the test digits, coloured by digit, are mixed: the position no longer tells the class. Right: one fixed $\mathbf{z}$ per row, the ten conditions along the row: the digit changes, slant and thickness stay. **c chooses what, z chooses how.**

Conditional VAE on MNIST



- each row: $\mathbf{c}$ = one digit, 0 to 9; each column: one random $\mathbf{z} \sim \mathcal{N}(0, I)$, the *same* in every row
- along a column the style stays and the digit changes: $\mathbf{z}$ carries the style, $\mathbf{c}$ the identity
- the generated digits are checked with a classifier of real digits

The Condition in a Convolutional Network



BloodMNIST: the one-hot cell type enters as 8 extra constant channels (1 for the requested type, 0 otherwise); stride-2 convolutions halve the size, transposed convolutions double it.

Layers of the Convolutional Conditional VAE

layer	output	param.
encoder, input: image and $\mathbf{c}$ *as channels,* $28 \times 28 \times (3+8)$		
conv 3×3, stride 2, ReLU	$14 \times 14 \times 32$	3 200
conv 3×3, stride 2, ReLU, flatten	$7 \times 7 \times 64$	18 496
dense $\to \boldsymbol{\mu}$	16	50 192
dense $\to \log \boldsymbol{\sigma}^2$	16	50 192
decoder, input $[\mathbf{z}, \mathbf{c}]$*:* $16 + 8$		
dense, ReLU, reshape	$7 \times 7 \times 64$	78 400
transposed conv 3×3, stride 2, ReLU	$14 \times 14 \times 32$	18 464
transposed conv 3×3, stride 2 (logits)	$28 \times 28 \times 3$	867

Shapes: height $\times$ width $\times$ channels; $\mathbf{c}$: the one-hot cell type (8 numbers); $d = 16$. A 3×3 convolution from C_{in} to C_{out} channels has $9\,C_{\text{in}}C_{\text{out}} + C_{\text{out}}$ parameters.

Total 219 811 parameters, all trainable (torchinfo).

Generating a Chosen Digit

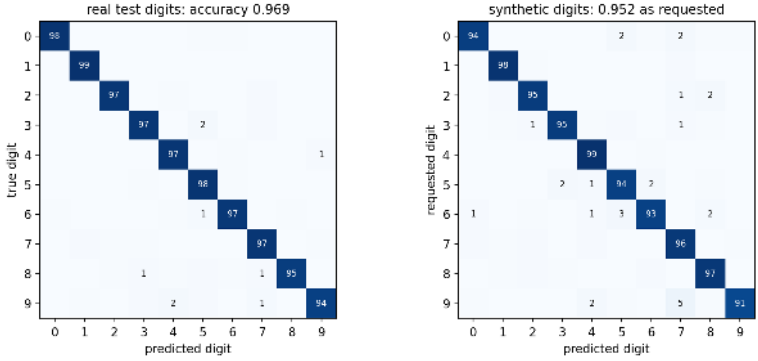
Only the decoder: condition $\mathbf{c} = \mathbf{e}_k$ (one-hot digit k), random style $\mathbf{z} \sim \mathcal{N}(0, I)$, image sigmoid $\big(G(\mathbf{z}, \mathbf{c})\big)$.

```
def generate_digit(k, n=10):
    c = F.one_hot(torch.full((n,), k),
                  10).float()
    z = torch.randn(n, cvae.d)
    with torch.no_grad():
        return torch.sigmoid(
            cvae.decode(z, c))
```

Row k: `generate_digit(k, 12)`; each column a new $\mathbf{z}$.

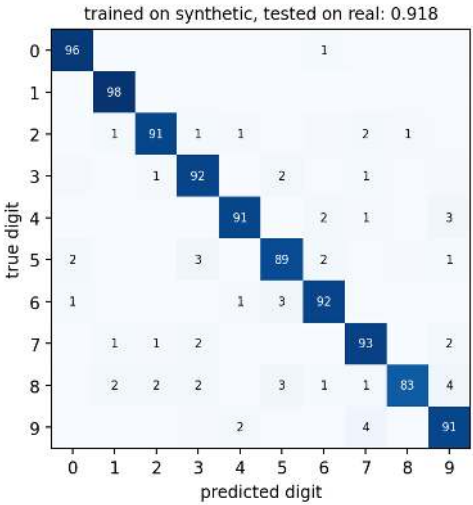


Classifier Check of Synthetic Digits



Classifier $784 \to 256 \to 10$, trained on real digits only. Left: on the real test digits (row: true digit). Right: on 1000 generated digits per class (row: requested digit); entries in % of the row. Diagonal mean: 0.95; the confusions resemble those of real digits.

Training on Synthetic Digits Only



The same classifier, trained on 60 000 *generated* digits (6000 per class, labels = requested digits), tested on the real test digits:

- trained on real digits: accuracy 0.969
- trained on synthetic digits: accuracy 0.918

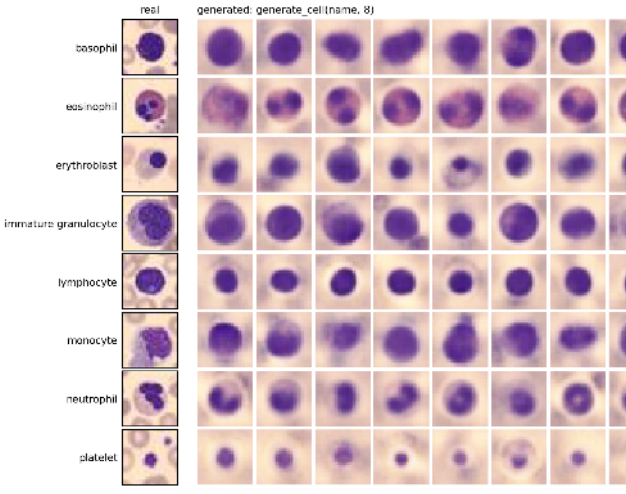
The gap measures what the generator misses of the real distribution: styles it never produces, and blurred strokes.

Generating a Cell of a Chosen Type

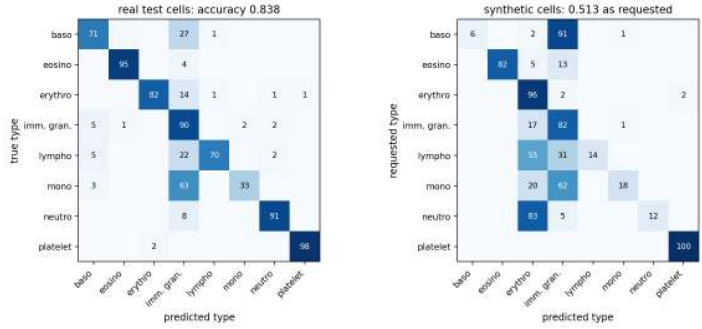
The same recipe, with the cell type as condition and an RGB output:

```
def generate_cell(name, n=8):
    k = names.index(name)
    c = F.one_hot(torch.full((n,),
                  8).float()
    z = torch.randn(n, bcvae.d)
    with torch.no_grad():
        return torch.sigmoid(
            bcvae.decode(z, c))
```

First column: one real cell of the type; then 8 generated.

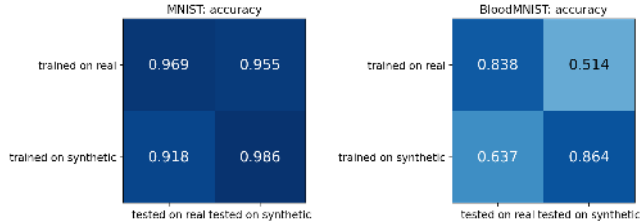


Classifier Check of Synthetic Cells



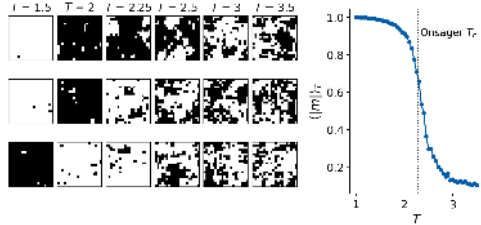
CNN trained on real cells only. Left: real test cells (accuracy 0.84). Right: 500 generated cells per requested type (diagonal mean 0.51). Platelets (100%) and erythroblasts (96%), told apart by size, come out right; generated basophils are taken for immature granulocytes (91%), neutrophils for erythroblasts (83%): the VAE blurs the shape of the nucleus.

Training and Testing on Real and Synthetic Data



- **trained on real, tested on synthetic:** does the generator draw the requested class?
- **trained on synthetic, tested on real:** do the synthetic data carry what is needed to recognise real instances?
- MNIST: synthetic digits almost replace real ones (0.92 against 0.97); blood cells: trained on synthetic, 0.64 on real cells (0.84 with real training) but 0.86 on synthetic: types easy to tell apart, but not the real ones

Ising Configurations at a Chosen Temperature



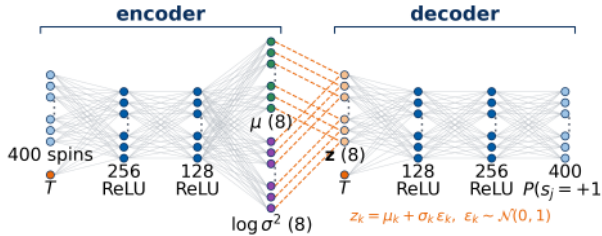
Monte Carlo configurations (white: up, black: down) and $\langle |m| \rangle_T$; dotted: Onsager's T_c

The 2D Ising model of Lecture 4, 20×20 spins $s_i = \pm 1$:

$$p(\mathbf{s} \mid T) = \frac{e^{-H(\mathbf{s})/T}}{Z(T)}, \quad H = -\sum_{\langle ij \rangle} s_i s_j$$

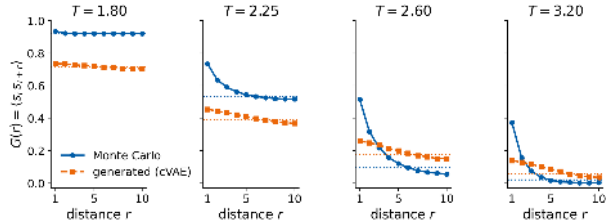
- data: 10 200 Metropolis configurations, 200 at each of 51 temperatures
- task: a conditional VAE generating configurations for a *requested* T
- test: does it reproduce the physics?

Ising Conditional VAE: Layers



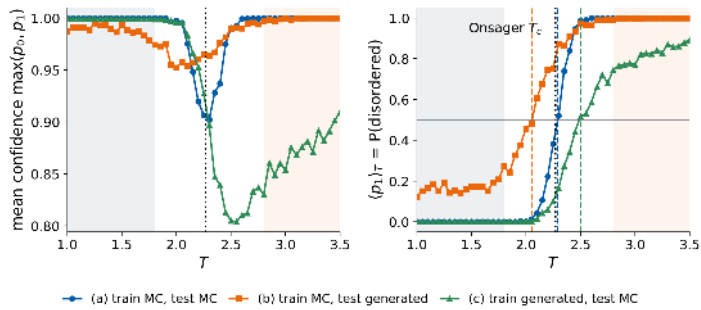
- pixels $x_i = (s_i + 1)/2 \in \{0, 1\}$, $D = 400$; condition $c = (T - 2.25)/0.75$ enters the encoder and the decoder
- encoder $401 \to 256 \to 128 \to (\mu, \log \sigma^2)$, latent dimension 8; decoder $9 \to 128 \to 256 \to 400$ Bernoulli logits
- parameters (torchinfo): encoder 137 872, decoder 137 104, total 274 976, all trainable

Generated Configurations against Monte Carlo



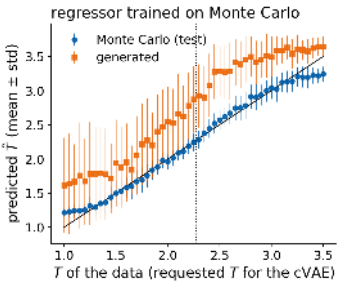
- correlation $G(r) = \langle s_i s_{i+r} \rangle$: generated spins are less correlated at short range and too correlated far apart (dotted: $\langle m \rangle^2$)
- $\langle |m| \rangle$: largest gap 0.234, at $T = 2.10$; energy per spin too high at *every* T, largest gap 0.634 at $T = 2.30$
- at $T \leq 1.5$, 8.5 % of generated configurations have $|m| < 0.5$ (Monte Carlo: none)

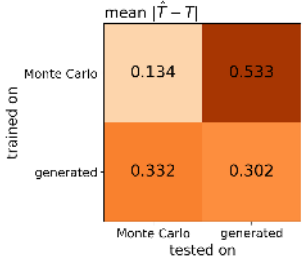
Physics Checks with a Phase Classifier



CNN classifier ordered/disordered, trained on $T < 1.8$ and $T > 2.8$ (shaded), T_c read where $\langle p_1 \rangle_T$ crosses $1/2$ (3 seeds): (a) MC $\to$ MC: 2.292 ± 0.012; (b) MC $\to$ generated: 2.056 ± 0.012; (c) generated $\to$ MC: 2.501 ± 0.057; Onsager: 2.269

Temperature Read Back from Generated Configurations





A CNN regressor predicts T from a configuration (mean absolute error $|\hat{T} - T|$, one seed per model).

- trained and tested on Monte Carlo: 0.134
- tested on generated: 0.533; they look *hotter* than requested, by up to 0.78 at $T = 2.5$

The cVAE learns the trend with T, not the exact Boltzmann distribution.

6

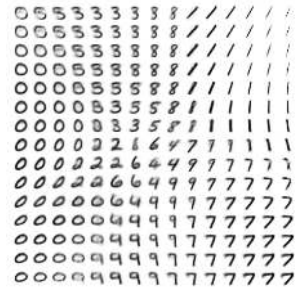
Disentangled Latent Features

JAGIELLONIAN UNIVERSITY
IN KRAKOW

Latent Coordinates and Features

Each latent coordinate z_k is a direction; moving along it, the others fixed, changes the decoded image (a traversal).

- in the $d = 2$ VAE (right) moving z_1 or z_2 changes several properties at once: the digit identity, its slant, its width
- **disentangled**: each coordinate responds to one independent property of the data and ignores the others
- the goal: one latent node $\leftrightarrow$ one distinct feature, found without labels (unlike the condition $\mathbf{c}$ of a cVAE)



$d = 2$ VAE: decoded images on a grid of (z_1, z_2)

The β-VAE Loss

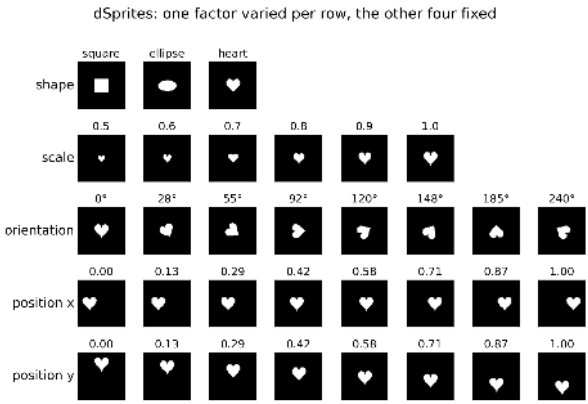
One weight more in the VAE loss:

$$\mathcal{L}_\beta(\mathbf{x}) = \underbrace{\mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}\big[-\log p_\theta(\mathbf{x}\,|\,\mathbf{z})\big]}_{\text{reconstruction}} + \beta\,\underbrace{\mathrm{KL}\big(q_\phi(\mathbf{z}\,|\,\mathbf{x})\,\|\,\mathcal{N}(0,I)\big)}_{\sum_k \frac{1}{2}(\mu_k^2+\sigma_k^2-\ln\sigma_k^2-1)}, \qquad \beta \geq 1$$

- $\beta = 1$: the VAE, minus the ELBO; $\beta > 1$: information stored in $\mathbf{z}$ costs more
- the prior $\mathcal{N}(0, I)$ has *independent* coordinates; a stronger pull towards it makes each kept coordinate carry one independent factor of variation
- a coordinate with $\mu_k \approx 0$, $\sigma_k \approx 1$ for every input (KL ≈ 0) carries nothing: the KL per coordinate shows which are used
- the price: blurrier reconstructions and samples; β trades sharpness for interpretable coordinates

I. Higgins et al., β-VAE, ICLR 2017, openreview.net/forum?id=Sy2fzU9gl

The dSprites Data



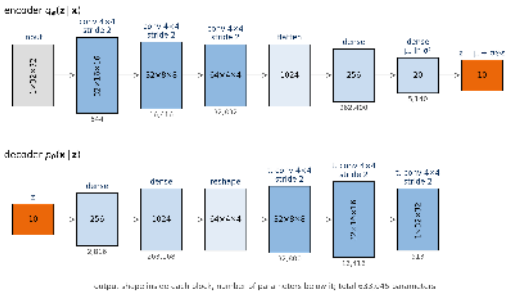
737 280 images of one sprite, every combination of five known factors once:

- shape (3), scale (6), orientation (40)
- position x and y (32 each)

Known factors: every latent coordinate can be checked against them. Here 100 000 training images at 32×32 pixels.

L. Matthey, I. Higgins, D. Hassabis, A. Lerchner, dSprites (2017), github.com/deepmind/dsprites-dataset, Apache 2.0

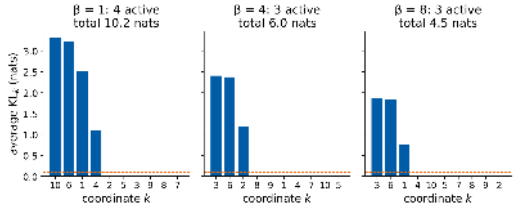
β-VAE Layers and Parameters



layer	output	param.
encoder, input $1 \times 32 \times 32$		
3 conv 4×4, stride 2	$64 \times 4 \times 4$	49 792
dense	256	262 400
dense $\to (\boldsymbol{\mu}, \log \boldsymbol{\sigma}^2)$	2×10	5 140
decoder, input $\mathbf{z}$*: 10*		
2 dense, reshape	$64 \times 4 \times 4$	265 984
3 transposed conv	$1 \times 32 \times 32$	49 729

dSprites, 32×32 pixels; shapes: channels $\times$ height $\times$ width; ReLU between layers. Total 633 045 parameters, all trainable (torchinfo).

Information per Latent Coordinate



Average KL_k of each of the 10 coordinates over the test images (dashed: 0.1 nats, the threshold for an *active* coordinate):

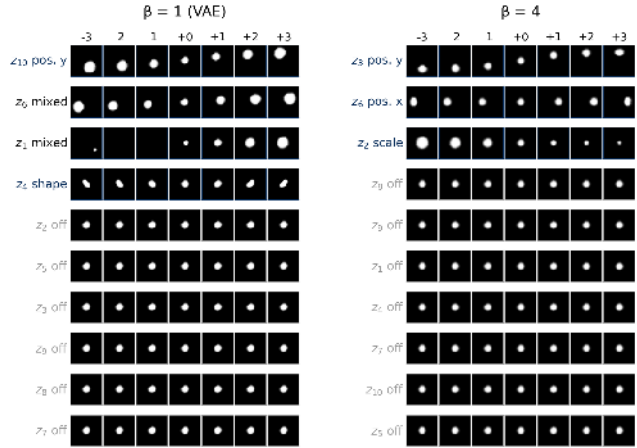
	$\beta = 1$	$\beta = 4$	$\beta = 8$
active coordinates	4	3	3
total KL (nats per image)	10.15	5.98	4.46
reconstruction (nats per image)	20.56	26.79	30.57

A larger β stores less information and reconstructs worse; the unused coordinates stay at the prior.

Sampling a β-VAE

Traversal: encode one image, move one coordinate z_k from -3 to $+3$, decode.

- $\beta = 4$: one coordinate each for position y, position x, scale; the rest off
- $\beta = 1$: position y separate, the rest mixed
- no labels: properties are found, then named by looking



Entropy of a Factor

dSprites: the true factors v_j of every image are known (shape, scale, orientation, position x, y). How uncertain is a factor before we look?

- surprise of a value b with probability $p(b)$: $-\ln p(b)$; 0 for a certain value, large for a rare one
- **entropy**: the average surprise

$$H(v) = -\sum_b p(b)\,\ln p(b)$$

- n equally likely values: $H = \ln n$, more values to tell apart $\Rightarrow$ larger entropy

factor	shape	scale	position x or y	orientation
values	3	6	32	40
H (nats)	1.10	1.79	3.47	3.69

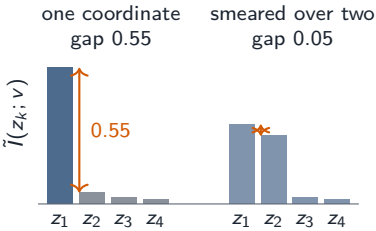
Mutual Information of a Coordinate and a Factor

How much uncertainty about factor v disappears once coordinate z_k is known?

$$I(z_k; v) = H(v) - H(v \mid z_k) = \sum_{a,b} p(a,b)\, \ln \frac{p(a,b)}{p(a)\,p(b)} = \mathrm{KL}\Big(p(a,b) \,\Big\|\, p(a)\,p(b)\Big)$$

- $H(v \mid z_k)$: uncertainty left after reading the coordinate
- $I = 0$: coordinate and factor independent; $I = H(v)$: the coordinate determines the factor
- normalised: $\tilde{I}(z_k; v) = I(z_k; v)/H(v) \in [0, 1]$, the fraction of the uncertainty removed
- detects any relation, also non-linear (an angle on a circle), unlike a correlation coefficient
- estimate: bin μ_k of the 10 000 test images into 20 bins; $p(a, b)$ = frequencies of (bin a, factor value b)

Mutual Information Gap



sketch for one factor v

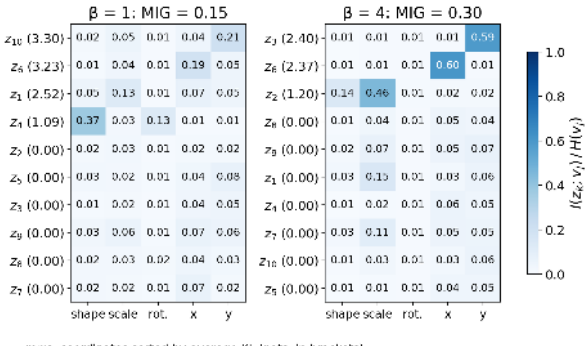
Gap between the best coordinate $z_{(1)}$ and the second $z_{(2)}$, averaged over the J factors:

$$\mathrm{MIG} = \frac{1}{J}\sum_{j=1}^{J}\Big[\tilde{I}(z_{(1)}; v_j) - \tilde{I}(z_{(2)}; v_j)\Big]$$

- large only if each factor sits in *one* coordinate
- factor copied into two (0.60, 0.60): gap 0
- needs the true factors: synthetic data, not galaxies

R. T. Q. Chen et al., NeurIPS 2018, arXiv:1802.04942

Measuring Disentanglement



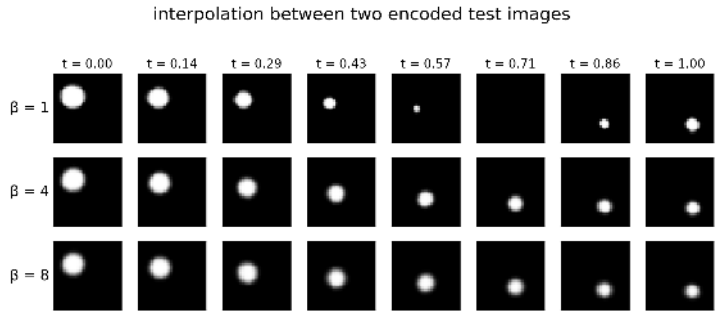
rows: coordinates sorted by average KL (nats, in brackets)

R. T. Q. Chen et al., NeurIPS 2018, arXiv:1802.04942

Matrix: $\tilde{I}(z_k; v_j) = I(z_k; v_j)/H(v_j)$ for every coordinate z_k (rows, by KL) and factor v_j (columns).

- MIG 0.15, 0.30, 0.30 for $\beta = 1, 4, 8$
- orientation: learnt by none; shape: weakly

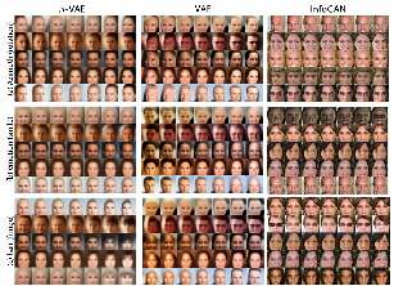
Interpolation between Two Images



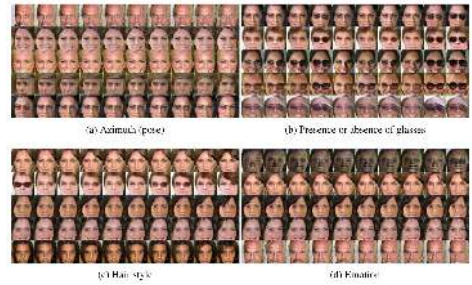
Encode two test images, decode the straight line $(1-t)\,\boldsymbol{\mu}_A + t\,\boldsymbol{\mu}_B$:

- $\beta = 1$: the sprite shrinks and vanishes on the way (empty canvas at $t = 0.71$) before reappearing
- $\beta = 4, 8$: the sprite moves and changes size smoothly, because position and size have their own coordinates

Disentanglement on Faces in the Literature



Higgins et al., ICLR 2017, Fig. 1: traversals on CelebA faces; columns β-VAE ($\beta = 250$), VAE, InfoGAN; rows: rotation, smile, fringe

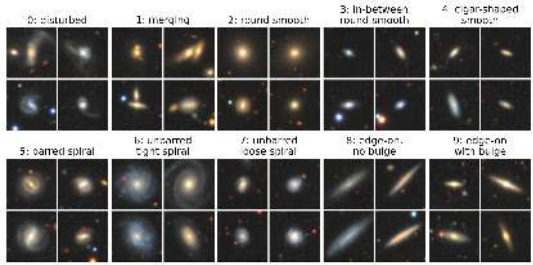


Chen et al., NIPS 2016, Fig. 6: InfoGAN codes for pose, glasses, hair style, emotion

On faces, rotation and smile get separate coordinates with large β; the plain VAE mixes rotation with smile, glasses and gender.

I. Higgins et al., ICLR 2017, openreview.net/forum?id=Sy2fzU9gl; X. Chen et al., NIPS 2016, arXiv:1606.03657

Galaxies: Galaxy10 DECaLS



- 17 736 colour images of galaxies (DESI Legacy Imaging Surveys, bands g, r, z), ten morphology classes from Galaxy Zoo volunteers
- central 128×128 pixels averaged to $64 \times 64 \times 3$; 15 076 for training, 2660 for testing

H. W. Leung, J. Bovy, Galaxy10 DECaLS, doi:10.5281/zenodo.10845026 (CC BY 4.0); M. Walmsley et al., MNRAS **509**, 3966 (2022); A. Dey et al., AJ **157**, 168 (2019)

Galaxies: the Question

A β-VAE is trained on the galaxy images *without labels*. Two questions:

1. does a latent coordinate encode a physical property of the galaxies, such as size, brightness, elongation or colour?
2. can such coordinates explain a classifier trained *with* the ten labels?

A latent coordinate has no name. That it means "size" is a **hypothesis**, tested in four stages:

1. *look*: traversals suggest what a coordinate changes
2. *measure*: properties computed from the pixels of every image, without labels
3. *test*: correlation of each coordinate with each property
4. *repeat*: a second training must give the same factor

Measured Properties of a Galaxy Image

Galaxy mask: connected pixels brighter than sky $+\,3\sigma_{\mathrm{sky}}$ around the centre; $f(\mathbf{r})$: sky-subtracted brightness inside the mask.

- size: area of the mask; brightness: $\sum_{\mathbf{r}} f(\mathbf{r})$; half-light radius r_{50}
- shape from the second moments of the light:

$$M_{ij} = \frac{\sum_{\mathbf{r}} f(\mathbf{r})\,(r_i - \bar{r}_i)(r_j - \bar{r}_j)}{\sum_{\mathbf{r}} f(\mathbf{r})}, \qquad \frac{b}{a} = \sqrt{\frac{\lambda_{\mathrm{min}}}{\lambda_{\mathrm{max}}}}, \qquad \tan 2\vartheta = \frac{2M_{xy}}{M_{xx} - M_{yy}}$$

  axis ratio b/a (1: round), orientation ϑ of the major axis
- ellipticity: $\varepsilon = \frac{1-b/a}{1+b/a}$, $e_1 = \varepsilon \cos 2\vartheta$, $e_2 = \varepsilon \sin 2\vartheta$
- colour $(R - B)/(R + B)$ of the channel sums; concentration r_{90}/r_{50}; offset of the light centre

Check: round smooth galaxies have median $b/a = 0.92$, edge-on disks 0.23.

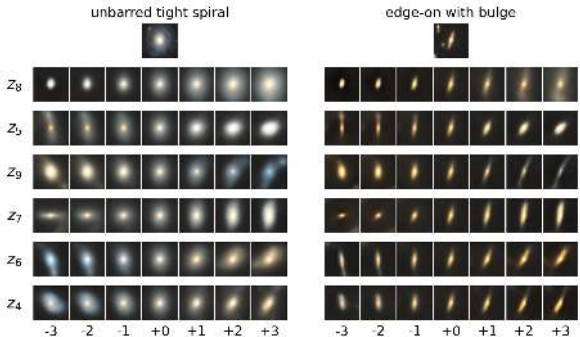
Naming a Latent Coordinate

- encode the 2660 test galaxies: centres $\mu_k(\mathbf{x}_i)$, $k = 1, \ldots, 10$
- correlation of coordinate k with property j over the test images:

$$R_{kj} = \left|\mathrm{corr}_i\Big(\mu_k(\mathbf{x}_i),\ \mathrm{proxy}_j(\mathbf{x}_i)\Big)\right|$$

- coordinate k is named after property j only if $R_{kj} \geq 0.5$ *and* R_{kj} exceeds its correlation with every other property by at least 0.15; otherwise it is "mixed" (size, brightness and r_{50} count as one family)
- repeat with a second model (another seed): its coordinates may come in another order and sign, so the partner of z_k is the coordinate with the largest $|\mathrm{corr}|$; a real factor has a partner near 1

Galaxy Traversals of a β-VAE

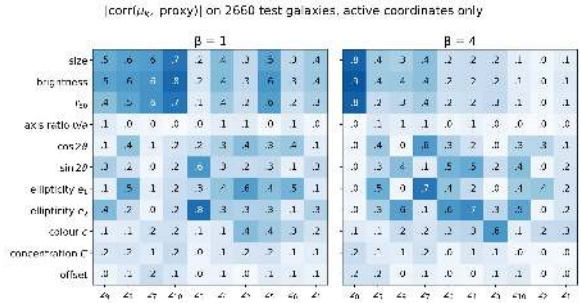


top: the real galaxy; rows: shift δ of one latent coordinate

$\beta = 4$, 10 latent coordinates, Gaussian decoder; each row moves one coordinate by δ from -3 to $+3$:

- z_8: a compact core grows into a large, bright disk
- z_7: elongation turns between horizontal and vertical
- z_5: a face-on spiral becomes an elongated disk
- no coordinate draws spiral arms or a bar: at 64 pixels they are lost

Latent Coordinates and Measured Proxies



Proxies measured from the pixels: size, brightness, axis ratio, ellipticity e_1, e_2, colour, concentration; shown is $|\mathrm{corr}(\mu_k, \text{proxy})|$.

- $\beta = 4$: z_8 follows brightness and size (0.85, 0.80); z_7: e_1 (0.72); z_3: colour (0.60)
- $\beta = 1$: six of ten coordinates mixed

Do the Factors Reappear?

- the same β-VAE trained a second time, from another random seed
- for every coordinate of the first model, find the coordinate of the second model with the largest $|\mathrm{corr}|$
- brightness and size reappear in a single coordinate: $|\mathrm{corr}| = 0.97$
- elongation and colour are spread differently: best partners only 0.53–0.79
- a factor seen in one training run is a hypothesis, not yet a finding

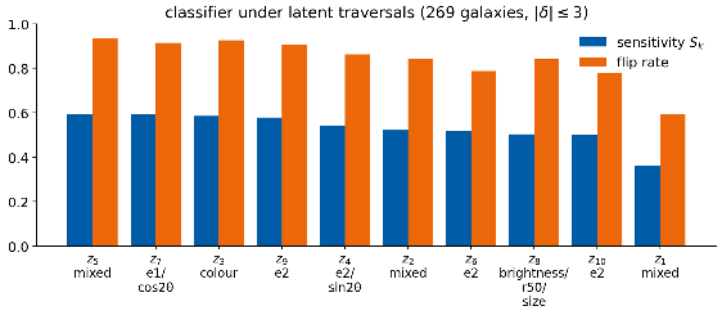
A Galaxy Classifier and Its Counterfactuals

- CNN classifier trained on the real images: test accuracy 0.764
- counterfactual of image $\mathbf{x}$ along coordinate k:

$$\mathbf{x}'(\delta) = G\Big(\boldsymbol{\mu}(\mathbf{x}) + \delta\, \mathbf{e}_k\Big)$$

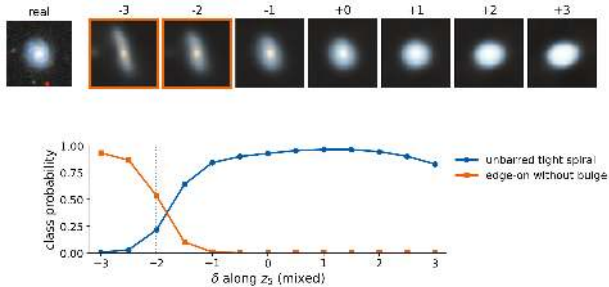
- check first: the reconstruction ($\delta = 0$) gets the class of the real image in only 42 % of the test galaxies
- only these 1129 faithful galaxies are used (269 of them analysed); otherwise a flip could come from the blurred reconstruction alone

Where the Class Flips



- the class flips along every coordinate (60–93 % of galaxies), typically already at $|\delta| = 1$
- a frequent flip: tight spiral $\rightarrow$ edge-on disk

One Counterfactual



- along z_5 the axis ratio drops from 0.89 to 0.37; at $\delta = -2$ "edge-on without bulge" overtakes "unbarred tight spiral"
- the classifier responds to elongation, as an astronomer would to a tilted disk; but the spiral arms are lost already at $\delta = 0$

Limits of Latent Interpretations

- without supervision, disentangled factors cannot be identified in general: many latent spaces explain the data equally well
- a coordinate gets a physical meaning only when it is checked against measured proxies and reappears across seeds
- counterfactuals explain a classifier only where the decoder reproduces the class of the real image
- at 64×64 pixels fine structure (spiral arms, bars) is not represented at all

F. Locatello et al., ICML 2019, arXiv:1811.12359

Galaxies: What a β-VAE Gives and What It Does Not

What we get

- one robust physical factor, found without labels: brightness and size (z_8, $|\mathrm{corr}| = 0.85, 0.80$), the same in a second training (0.97)
- candidate factors for elongation (0.72) and colour (0.60)
- counterfactuals that test a classifier: tilting a spiral towards edge-on (axis ratio $0.89 \to 0.37$) changes its class, as it should

What we do not get

- a guarantee: elongation and colour are spread differently in the second training (0.53–0.79); with $\beta = 1$ six coordinates mix
- spiral arms and bars: lost at 64×64 pixels
- faithful reconstructions for most galaxies: only 42 % keep their class, so counterfactuals hold only for those

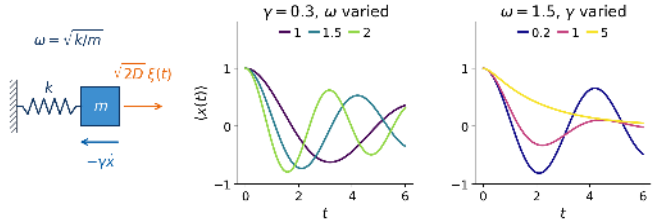
latent factors are *hypotheses*: check them against measured proxies and across trainings

Physical Parameters from Trajectories

- a physicist compresses observations into a few parameters and answers questions from them (Iten et al.)
- galaxies: the true causes are unknown, latent factors are tested against measured proxies
- trajectories: simulated, so the true physical parameters of every sample are *known*
- the same test as before, with the answer in hand
  1. a β-VAE on noisy oscillators and the parameters it finds
  2. SciNet of Iten et al.: a network that answers questions
  3. two coupled oscillators: springs or normal modes

R. Iten, T. Metger, H. Wilming, L. del Rio, R. Renner, Phys. Rev. Lett. **124**, 010508 (2020), doi:10.1103/PhysRevLett.124.010508

The Noisy Damped Oscillator

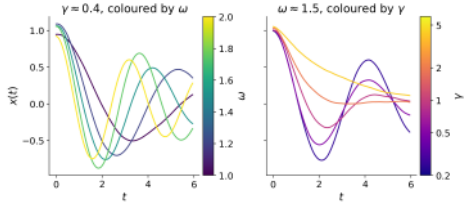


Mass on a spring in a fluid, kicked at random (Langevin equation):

$$\ddot{x} = -\omega^2 x - \gamma\dot{x} + \sqrt{2D}\,\xi(t), \qquad \langle \xi(t)\,\xi(t') \rangle = \delta(t - t')$$

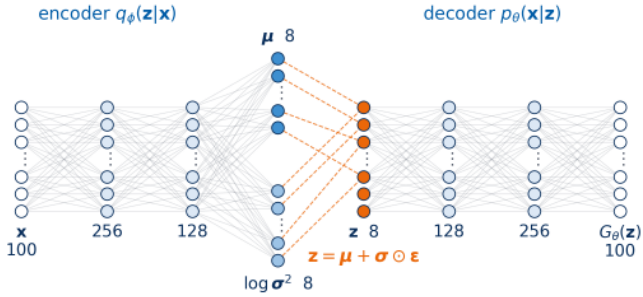
- ω: frequency; γ: friction rate; ξ: white noise of strength D
- $\gamma > 2\omega$: overdamped, no oscillation
- simulated with the Euler–Maruyama scheme, $\Delta t = 0.01$

Trajectory Data



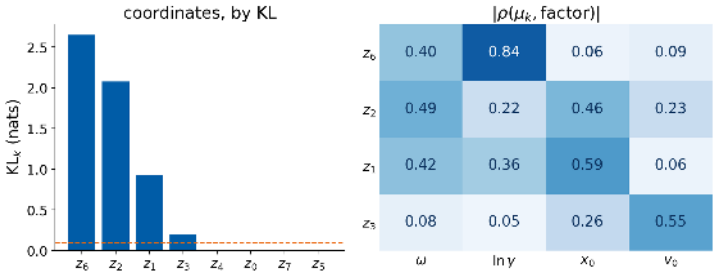
- one sample $\mathbf{x}_i \in \mathbb{R}^{100}$: positions at $t_j = 0.06\,j$, from 0 to 5.94; 20 000 for training, 5000 for testing
- four factors per trajectory: $\omega \sim \mathcal{U}[1, 2]$, $\ln \gamma \sim \mathcal{U}[\ln 0.2, \ln 6]$ (*physical*); $x_0 \sim \mathcal{U}[0.5, 1.5]$, $v_0 \sim \mathcal{U}[-0.5, 0.5]$ (*nuisance*); kept for evaluation only
- noise $D = 0.002$: rms deviation from the mean trajectory 0.0315; 20.9 % of the trajectories overdamped

β-VAE for Trajectories



- dense layers $100 \to 256 \to 128 \to 8$ and back; 120 692 parameters (torchinfo)
- Gaussian decoder, width $\sigma_x = 0.1$: $\|\mathbf{x} - G_\theta(\mathbf{z})\|^2/(2\sigma_x^2) + \beta\,\mathrm{KL}$
- $\beta = 0.5, 1, 4, 8$, three seeds each: 12 models

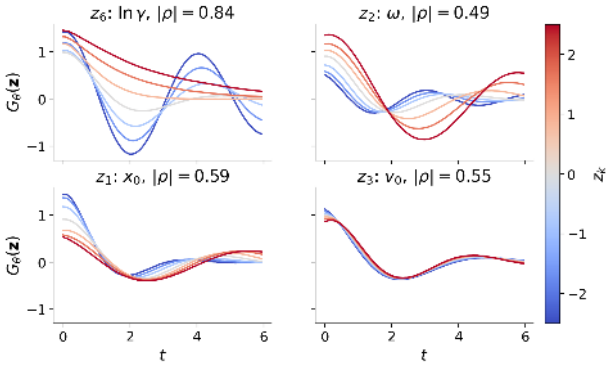
Latent Coordinates of the Oscillator



$\beta = 4$, seed 0: four active coordinates for four factors

- z_6 (2.65 nats) follows the damping: $|\rho| = 0.84$ with $\ln\gamma$
- z_2 mixes frequency and initial position (0.49 and 0.46)
- z_1: initial position (0.59); z_3: initial velocity (0.55)

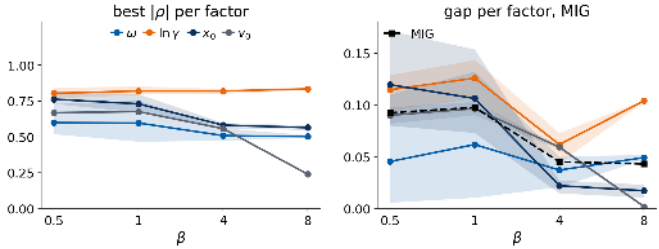
Traversals of the Oscillator Coordinates



Decoded mean trajectory $G_\theta(\mathbf{z})$ while one coordinate moves from -2 to 2:

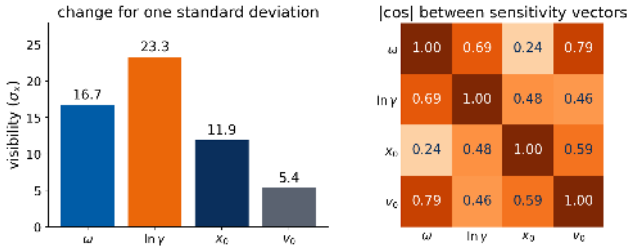
- z_6: from an oscillation to a slow creep, the damping
- z_2: the period changes together with the start
- z_1: the initial position
- z_3: a small change of the initial slope

Disentangling against β



- damping found for every β: best $|\rho| = 0.80$–0.83 (mean over seeds)
- MIG 0.092, 0.098, 0.045, 0.043 for $\beta = 0.5, 1, 4, 8$; the true factors give 0.916, ω and x_0 rotated by 45° give 0.462
- larger β does *not* raise the MIG here; at $\beta = 8$ the initial velocity is dropped

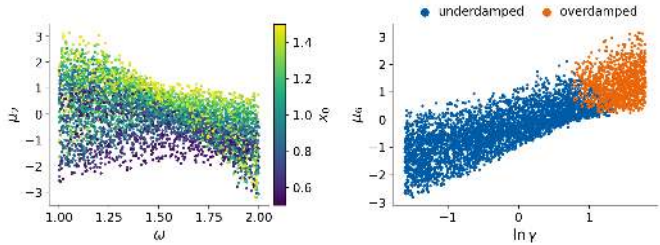
What the Trajectories Allow



Fisher information of the mean trajectory $\langle \mathbf{x} \rangle(\mathbf{p})$: $F_{ab} = \sigma_x^{-2} \sum_i \partial_a \langle x(t_i) \rangle \, \partial_b \langle x(t_i) \rangle$

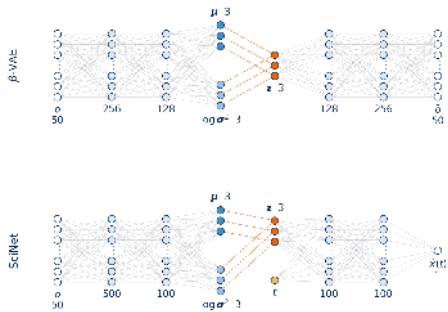
- visibility $\sqrt{F_{aa}}\,\mathrm{std}(p_a)$: change of the trajectory in units of σ_x for one standard deviation of p_a
- $|\cos|$ between sensitivity directions: ω–v_0 0.79, ω–$\ln\gamma$ 0.69: changes of ω are partly mimicked by other factors

Frequency and Damping Coordinates



- frequency coordinate: $R^2 = 0.241$ with ω alone, 0.449 with ω and x_0, 0.969 with all factors, squares and products
- overdamped trajectories decay at the slow rate $\approx \omega^2/\gamma$: there z_6 follows $\ln(\omega^2/\gamma)$ (0.66), not $\ln\gamma$ (0.12)
- the network stores the combination that the data contain

SciNet: a Network That Answers Questions



- encoder: 50 observed positions $\to$ 3 latent neurons
- question: a time $t \in [0, 10]$, appended to $\mathbf{z}$
- decoder: one answer, $\hat{x}(t)$
- cost $(\hat{x} - x)^2 + \beta\,\mathrm{KL}$, $\beta = 10^{-3}$
- ELU layers as in the paper; 86 907 parameters

Pendulum $\ddot{x} = -\kappa x - b\dot{x}$, $\kappa \in [5, 10]$, $b \in [0.5, 1]$, fixed start, no noise

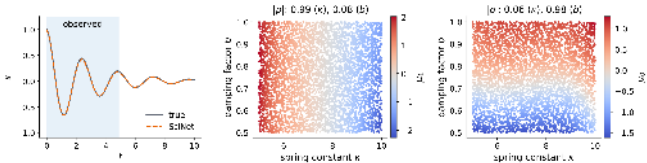
R. Iten et al., Phys. Rev. Lett. **124**, 010508 (2020), arXiv:1807.10300

SciNet and β-VAE Compared

	β-**VAE**	**SciNet**
input	observed trajectory	observed trajectory and a time t
output	the same trajectory	one number, $\hat{x}(t)$
latent space holds	all that rebuilds the input	what answers the questions
prediction after the window	no	yes, up to $t = 10$
loss	$\|\mathbf{x} - G\|^2/(2\sigma_x^2) + \beta\,\mathrm{KL}$	$(\hat{x} - x)^2 + \beta\,\mathrm{KL}$

- an autoencoder is SciNet with the single question "repeat the input" (Iten et al., Corollary 4)
- the two β are on different scales: times $2\sigma_x^2 = 0.02$, our KL weight is $0.02\,\beta$ against all squared errors, about 10^{-3} per point at $\beta = 4$

SciNet on the Pendulum of Iten et al.

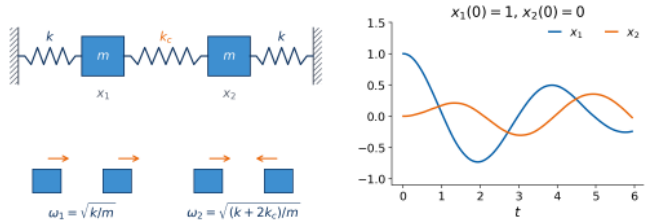


SciNet, 1000 epochs: prediction from the observed window, and the two active neurons over (κ, b)

data	network	best $\lvert\rho\rvert$: κ, b	rms error
paper setting	SciNet, 1000 epochs	0.99, 0.98	0.0178
paper setting	β-VAE	0.99, 0.97	—
with noise	SciNet, β-VAE	0.99, 0.88 | 0.98, 0.95	0.0609
noise, random x_0, v_0	SciNet, β-VAE	0.93, 0.31 | 0.90, 0.27	0.0725

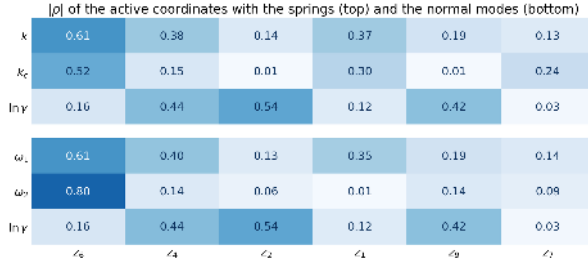
Both networks separate κ and b on the paper's data; with random initial conditions both lose the damping.

Two Coupled Oscillators



- normal modes: in phase, $\omega_1 = \sqrt{k/m}$; anti-phase, $\omega_2 = \sqrt{(k+2k_c)/m}$
- a trajectory depends on the springs only through ω_1, ω_2
- k, k_c drawn independently, so ω_1, ω_2 are correlated (0.63), and the two descriptions can be told apart

Springs or Normal Modes



- $\beta = 4$, two seeds, six active coordinates for seven factors
- best $|\rho|$ with ω_2: 0.80 and 0.79; with k_c: only 0.52 and 0.59
- the coupling is stored through the normal-mode frequency ω_2; ω_1 gets no coordinate of its own

Trajectories: What a β-VAE Gives and What It Does Not

What we get

- damping in its own coordinate for every β (0.80–0.83)
- overdamped: the combination ω^2/γ the data contain
- clean pendulum: κ and b separated by both networks
- coupled masses: the mode frequency ω_2

What we do not get

- a clean frequency coordinate: ω mixed with x_0
- a high MIG: 0.04–0.13, against 0.916 for the true factors
- robustness to random x_0, v_0: both networks lose the damping

what a network can separate is set by the data: how visible the parameters are and how much their effects overlap

VAE, Conditional VAE and β-VAE

	VAE	conditional VAE	β-VAE
trained on	images $\mathbf{x}$	pairs $(\mathbf{x}, \mathbf{c})$	images $\mathbf{x}$
loss	reconstruction + KL	reconstruction + KL	reconstruction + β KL
to sample	$\mathbf{z} \sim \mathcal{N}(0, I)$	choose $\mathbf{c}$, $\mathbf{z} \sim \mathcal{N}(0, I)$	$\mathbf{z} \sim \mathcal{N}(0, I)$, or move one z_k
$\mathbf{z}$ controls	everything, mixed	what $\mathbf{c}$ does not say (style)	aims at one property per coordinate
labels	not needed	needed	not needed
use	new samples	samples of a chosen class or T	finding and controlling factors

Summary

- a **generative model** learns the distribution $p(\mathbf{x})$ of the data from samples, and draws new samples from it
- **maximum likelihood**: choose the parameters that make the observed data most probable; it minimises the KL divergence to the data
- **VAE**: a latent space with a known distribution, $\mathcal{N}(0, I)$; trained on reconstruction + KL; decode random $\mathbf{z}$ to generate
- **conditional VAE**: generates what we ask for, a digit or a cell type
- **β-VAE**: pushes each latent coordinate towards one factor, without labels; galaxies and trajectories show that each factor is a hypothesis, tested against measured or known parameters and across trainings

Next Lecture: Decision Trees and Ensembles

- data as tables: one row per sample, one column per measured feature
- decision trees: splits chosen by the Gini impurity or the entropy, rules that can be read
- random forests: many trees grown on resampled data, their votes combined
- boosting: trees added one at a time, each correcting the errors of the previous ones

Thank You!

Questions, project ideas, feedback: marcin.plodzien@uj.edu.pl

Website · GitHub · Google Scholar



